

The Covering Excess Method in the Theory of Covering Codes

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Chapter 1

Introduction

Let $K(n, R)$ denote the minimal cardinality of a binary code of length n and covering radius R . This thesis is concerned with lower bounds on $K(n, R)$ by the method of covering excess, which was first proposed by Johnson [18] in 1972 and later rediscovered by van Wee [29] in 1988.

1.1 A sphere-covering problem

Let $\mathbb{F} = \{0, 1\}$ denote the finite field with two elements. To measure the similarity between two words of length n over $\mathbb{F}$, we introduce their Hamming distance, which simply counts the number of places, where the words differ. The space $\mathbb{F}^n$ of the binary words of length n over $\mathbb{F}$, equipped with the Hamming distance, is one of the most important finite metric spaces.

Like in every (finite) metric space there are two fundamental questions, the sphere-packing and the sphere-covering problem.

- (i) Determine the maximal number $A(n, 2e+1)$ of non-overlapping spheres of given radius e , that can be placed in the Hamming space $\mathbb{F}^n$.
- (ii) Determine the minimal number $K(n, R)$ of spheres of given radius R needed to cover the whole space $\mathbb{F}^n$.

The quantity $A(n, d)$ was widely studied during the last six decades in the framework of error-correcting codes. The sphere-covering problem (ii) however was not studied equally extensive, but in the last two or three decades, it attracted much attention to the mathematicians. In generally the determination of $K(n, R)$ is very difficult and we have to be content with lower

and upper bounds. A trivial lower bound for $K(n, R)$ certainly is the cardinality of the whole space $\mathbb{F}^n$ divided by the cardinality of a Hamming sphere with radius R . Many methods are known to improve on this lower bound, the so-called sphere-covering bound. One of the most efficient method is the covering excess counting method, which is the object of the present thesis. It relies on the fact, that for arithmetical reasons under certain conditions a sphere-covering is forced to cover words in a small Hamming sphere more than once, the so-called covering excess.

1.2 Overview of the thesis

In chapter 2 we give the necessary background on codes as well as the notations and terminology on covering excess, which are used throughout this work. As a first instance of the covering excess counting method we give Johnson's and van Wee's bound on $K(n, 1)$ for even n , which makes use of a forced covering excess in a sphere of radius one.

In chapter 3 we study bounds for $K(n, 1)$ when n is odd. We first use a method of Blass and Litsyn [1] to give the presently best known lower bound for $K(n, 1)$ if $n \equiv 5 \pmod{6}$. Then we discuss the general excess bound, which follows from the fact, that if $n \equiv -1 \pmod{p}$ for an odd prime p , there exists a covering excess on a sphere with radius $p - 1$. We give a new lower bound for the covering excess, leading to a "general improvement of the general excess bound" for binary codes with covering radius one.

In chapter 4 we study bounds for $K(n, R)$, which rely on a covering excess in spheres with radius one. We describe a classical bound of van Wee for this case from 1988 and a slight improvement of this bound by Honkala. We present a short proof of a further improvement of Honkala's bound. Moreover we give an improvement of the van Wee bound in case that $R + 1$ divides n , when both bounds coincide and no general further progress of the van Wee bound is known up to now.

Chapter 2

Preliminaries

2.1 Codes

Let $\mathbb{F} = \{0, 1\}$ denote the finite field with two elements. The Hamming distance $d(\mathbf{x}, \mathbf{y})$ between $\mathbf{x} = (x_1, \dots, x_n) \in \mathbb{F}^n$ and $\mathbf{y} = (y_1, \dots, y_n) \in \mathbb{F}^n$ is defined by

$$d(\mathbf{x}, \mathbf{y}) = |\{i \in \{1, \dots, n\} \mid x_i \neq y_i\}|.$$

One easily checks, that $\mathbb{F}^n$, equipped with the Hamming distance, is a finite metric space, the so-called Hamming space.

Any nonempty subset $C \subset \mathbb{F}^n$ is called a code. For $\mathbf{x} \in \mathbb{F}^n$ the distance $d(\mathbf{x}, C)$ is defined by

$$d(\mathbf{x}, C) = \min_{\mathbf{c} \in C} d(\mathbf{x}, \mathbf{c}).$$

We say that $\mathbf{x}$ R -covers $\mathbf{y}$, if $d(\mathbf{x}, \mathbf{y}) \leq R$ holds. C R -covers $\mathbf{x}$ means $d(\mathbf{x}, C) \leq R$. In both cases we simply say "cover" instead of " R -cover", if R is clear from the context.

If $|C| \geq 2$, then $d(C)$ denotes the minimum distance of C :

$$d(C) = \min_{\mathbf{c}_1, \mathbf{c}_2 \in C, \mathbf{c}_1 \neq \mathbf{c}_2} d(\mathbf{c}_1, \mathbf{c}_2).$$

Let $B(\mathbf{x}, e)$ denote the Hamming sphere with radius e centered on $\mathbf{x} \in \mathbb{F}^n$,

$$B(\mathbf{x}, e) = \{\mathbf{y} \in \mathbb{F}^n : d(\mathbf{x}, \mathbf{y}) \leq e\}.$$

One easily sees, that the cardinality of a sphere $B(\mathbf{x}, e)$ does not depend on

$\mathbf{x} \in \mathbb{F}^n$. So we may set

$$V(n, e) = |B(\mathbf{x}, e)| = \sum_{0 \leq i \leq e} \binom{n}{i} \quad (2.1)$$

for any $\mathbf{x} \in \mathbb{F}^n$.

Like in every (finite) metric space there exists the fundamental sphere-packing problem:

- (i) Determine the maximal number of non-overlapping spheres of given radius e , that can be placed in the Hamming space $\mathbb{F}^n$.

One easily sees, that the centers of the spheres in such a packing form a code $C \subset \mathbb{F}^n$ with minimum distance at least $2e + 1$. Thus the number requested in (i) is $A(n, 2e + 1)$, where $A(n, d)$ denotes the maximal cardinality of a code $C \subset \mathbb{F}^n$ with minimum distance at least d .

Assume that a set of messages is encoded by the words of a code $C \subset \mathbb{F}^n$ with minimum distance at least $2e + 1$. If during the transmission of such a word there occur at most e errors, then these errors can be corrected by the receiver. Thus $A(n, d)$ is the most basic quantity in coding theory, since $A(n, 2e + 1)$ is the maximal cardinality of a code correcting up to e errors. For a monograph on error-correcting codes see [26]. Updated internet tables for $A(n, d)$ are given by Brouwer [2].

In this thesis however we are concerned with the dual sphere-covering problem:

- (ii) Determine the minimal number of spheres of given radius R needed to cover the whole space $\mathbb{F}^n$.

The covering radius $R = R(C)$ of a code $C \subset \mathbb{F}^n$ denotes the least integer R , such that the spheres with radius R centered on the codewords cover the whole space. It is clear, that the centers of the spheres of a covering described in (ii) form a code with covering radius at most R . Equivalently, a code $C \subset \mathbb{F}^n$ has covering radius (at most) R , if

$$\forall \mathbf{x} \in \mathbb{F}^n \exists \mathbf{c} \in C \quad \text{with} \quad d(\mathbf{x}, \mathbf{c}) \leq R \quad (2.2)$$

holds or, simply, if each $\mathbf{x} \in \mathbb{F}^n$ is R -covered by C . $K(n, R)$ denotes the minimal cardinality of a code $C \subset \mathbb{F}^n$ with covering radius (at most) R .

In the last two or three decades the theory of covering codes attracted a great amount of attention. A monograph [4] appeared in 1997. Updated internet tables for $K(n, R)$ are given by Kéri [19]. An internet bibliography on covering codes is compiled by Lobstein [25].

Let $\mathbf{0} = (0, \dots, 0) \in \mathbb{F}^n$ denote the all-zero word of length n . Replacing the i th coordinate of the all-zero word by one yields the unit vector $\mathbf{e}_i \in \mathbb{F}^n$, $1 \leq i \leq n$. By the weight $w(\mathbf{x})$ of a word $\mathbf{x} \in \mathbb{F}^n$ we denote the number of its nonzero coordinates. The support of $\mathbf{x} \in \mathbb{F}^n = (x_1, \dots, x_n)$, $\text{supp}(\mathbf{x})$, is defined by

$$\text{supp}(\mathbf{x}) = \{1 \leq i \leq n \mid x_i \neq 0\}. \quad (2.3)$$

Binary vectors may be identified with their supports and if $\mathbf{x}, \mathbf{y} \in \mathbb{F}^n$, then $\mathbf{x} \subset \mathbf{y}$, $\mathbf{x} \cup \mathbf{y}$ and $\mathbf{x} \cap \mathbf{y}$ refer to the supports of $\mathbf{x}$ and $\mathbf{y}$.

We further set

$$\mathbb{F}_i^n = \{\mathbf{x} \in \mathbb{F}^n \mid \text{the } i\text{th coordinate of } \mathbf{x} \text{ equals } 1\}.$$

Let $C \subset \mathbb{F}^n$ be a code with covering radius R and cardinality $K(n, R)$. Then by $\mathbb{F}^n \subset \bigcup_{\mathbf{c} \in C} B(\mathbf{c}, R)$ we get $|\mathbb{F}^n| \leq \sum_{\mathbf{c} \in C} |B(\mathbf{c}, R)| = |C|V(n, R)$ and thus

$$K(n, R) \geq \frac{2^n}{V(n, R)}. \quad (2.4)$$

This is the most basic lower bound on $K(n, R)$, called the sphere-covering bound. In generally this bound cannot be improved, since there are codes attaining equality in (2.4). These codes are called perfect. One especially famous family of perfect codes are the so-called Hamming codes.

Theorem 1 (Hamming [11]). *If*

$$n = 2^m - 1,$$

then there exists a perfect code $C \subset \mathbb{F}^n$ with covering radius one. Especially

$$K(n, 1) = 2^{n-m}.$$

In the majority of cases however the sphere-covering bound can be improved. Many different methods are known to do this, see [4]. One of the most efficient method is the covering excess counting method, which is the object of the present thesis. It relies on the fact, that sometimes a covering code is forced to cover words in a small Hamming sphere more than once, the

so-called covering excess. As a simple example it could be possible, that a codeword covers always an even number of points in a Hamming sphere, but the cardinality of the sphere is odd. Then a covering excess in that sphere is unavoidable.

The first who introduced the method of covering excess was Johnson [18] in 1972, but his remarkable paper remained unnoticed until the appearance of a monograph on covering codes [4] in 1997. The method was rediscovered by van Wee [29] in 1988, so it should be quite fair, to attribute the method both to Johnson and van Wee. After some prerequisites in section 2.2 we recall their most prominent example in section 2.3, a lower bound for $K(n, 1)$ for even n .

2.2 The covering excess

In this section we give the definition of covering excess, some notations and a few basic equations, which will be used throughout this work.

Let $C \subset \mathbb{F}^n$ be a code with covering radius R . For $\mathbf{x} \in \mathbb{F}^n$ we define the covering excess $E(\mathbf{x})$ by

$$E(\mathbf{x}) = E_C(\mathbf{x}) = |B(\mathbf{x}, R) \cap C| - 1. \quad (2.5)$$

By (2.2) $E(\mathbf{x}) \geq 0$ for every $\mathbf{x} \in \mathbb{F}^n$, and $E(\mathbf{x})$ is strictly positive, if $\mathbf{x}$ is covered by at least two codewords from C . For $V \subset \mathbb{F}^n$, $i \geq 0$ and $\mathbf{x} \in \mathbb{F}^n$ we set

$$\begin{aligned} E(V) &= \sum_{\mathbf{y} \in V} E(\mathbf{y}), \\ E_i(\mathbf{x}) &= \sum_{\mathbf{y} \in \mathbb{F}^n, d(\mathbf{x}, \mathbf{y})=i} E(\mathbf{y}), \\ Z &= \{\mathbf{z} \in \mathbb{F}^n \mid E(\mathbf{z}) > 0\}, \\ Z_i &= \{\mathbf{z} \in \mathbb{F}^n \mid E(\mathbf{z}) = i\}. \end{aligned}$$

For $V \subset \mathbb{F}^n$ we have

$$\begin{aligned}
E(V) &= \sum_{\mathbf{x} \in V} (|B(\mathbf{x}, R) \cap C| - 1) \\
&= \sum_{\mathbf{x} \in V} \sum_{\mathbf{c} \in B(\mathbf{x}, R) \cap C} 1 - |V| \\
&= \sum_{\mathbf{c} \in C} \sum_{\mathbf{x} \in B(\mathbf{c}, R) \cap V} 1 - |V| \\
&= \sum_{\mathbf{c} \in C} |B(\mathbf{c}, R) \cap V| - |V|.
\end{aligned} \tag{2.6}$$

In the special case $V = \mathbb{F}^n$ we get

$$E(\mathbb{F}^n) = |C|V(n, R) - 2^n, \tag{2.7}$$

which in case of $R = 1$ by (2.1) further simplifies to

$$E(\mathbb{F}^n) = |C|(n + 1) - 2^n. \tag{2.8}$$

2.3 A bound for $K(n, 1)$ with even n

The case of binary codes with covering radius one is of special interest. Here the sphere-covering bound (2.4) reads

$$K(n, 1) \geq \frac{2^n}{n + 1}. \tag{2.9}$$

We now present the argument of Johnson and van Wee to improve on this bound for even n , which is probably the most famous application of the covering excess method.

Theorem 2 (Johnson [18], van Wee [29]). *Let $C \subset \mathbb{F}^n$ be a binary code with covering radius one. If $\mathbf{x} \in \mathbb{F}^n$, then*

$$E(B(\mathbf{x}, 1)) \equiv \begin{cases} 1 \pmod{2} & \text{if } n \text{ is even and } \mathbf{x} \notin C, \\ 0 \pmod{2} & \text{if } n \text{ is odd.} \end{cases}$$

Proof. Let $\delta(\mathbf{x}) = 1$ if $\mathbf{x} \in C$, otherwise $\delta(\mathbf{x}) = 0$. If $\mathbf{c} \in C$, then $|B(\mathbf{c}, 1) \cap B(\mathbf{x}, 1)|$ equals the number of words in the sphere $B(\mathbf{x}, 1)$, which were 1-covered by $\mathbf{c}$. One easily sees, that

$$|B(\mathbf{c}, 1) \cap B(\mathbf{x}, 1)| = \begin{cases} |B(\mathbf{x}, 1)| = n + 1 & \text{if } d(\mathbf{c}, \mathbf{x}) = 0, \\ 2 & \text{if } d(\mathbf{c}, \mathbf{x}) = 1 \text{ or } 2, \\ 0 & \text{if } d(\mathbf{c}, \mathbf{x}) \geq 3. \end{cases}$$

Now (2.6) applied to $V = B(\mathbf{x}, 1)$ yields

$$\begin{aligned} E(B(\mathbf{x}, 1)) &= \sum_{\mathbf{c} \in C} |B(\mathbf{c}, 1) \cap B(\mathbf{x}, 1)| - |B(\mathbf{x}, 1)| \\ &= \delta(\mathbf{x})(n + 1) + 2 \sum_{\mathbf{c} \in C, 1 \leq d(\mathbf{c}, \mathbf{x}) \leq 2} 1 - (n + 1) \\ &\equiv (\delta(\mathbf{x}) - 1)(n + 1) \pmod{2} \end{aligned}$$

and Theorem 2 follows. $\square$

Now let n be even and $C \subset \mathbb{F}^n$ a binary code with covering radius one. Since $E(B(\mathbf{x}, 1))$ is always nonnegative, from Theorem 2 follows

$$E(B(\mathbf{x}, 1)) \geq 1 \quad \text{if } \mathbf{x} \notin C. \quad (2.10)$$

Thus there exists a covering excess on every sphere of radius one not centered on a codeword. This was used by Johnson and van Wee to derive the following improvement of the sphere-covering bound (2.9), the first excess bound for binary codes with covering radius one (see [4]).

Theorem 3 (Johnson [18], van Wee [29]). *When n is even we have*

$$K(n, 1) \geq \frac{2^n}{n}.$$

Proof. Let $C \subset \mathbb{F}^n$ be a binary code with covering radius one. Note, that $\mathbf{z} \in Z$ means $E(\mathbf{z}) \geq 1$ and thus $|B(\mathbf{z}, 1) \cap C| \geq 2$, implying

$$|B(\mathbf{z}, 1) \setminus C| \leq |B(\mathbf{z}, 1)| - 2 = n - 1 \quad \text{if } \mathbf{z} \in Z. \quad (2.11)$$

We now get

$$\begin{aligned}
2^n - |C| &= \sum_{\mathbf{x} \in \mathbb{F}^n \setminus C} 1 \\
&\leq \sum_{\mathbf{x} \in \mathbb{F}^n \setminus C} E(B(\mathbf{x}, 1)) \quad \text{by (2.10)} \\
&= \sum_{\mathbf{x} \in \mathbb{F}^n \setminus C} \sum_{\mathbf{z} \in B(\mathbf{x}, 1)} E(\mathbf{z}) \\
&= \sum_{\mathbf{z} \in Z} E(\mathbf{z}) \sum_{\mathbf{x} \in B(\mathbf{z}, 1) \setminus C} 1 \\
&\leq (n-1)E(\mathbb{F}^n) \quad \text{by (2.11)} \\
&= (n-1)(|C|(n+1) - 2^n) \quad \text{by (2.8)}
\end{aligned}$$

implying $|C| \geq 2^n/n$ and Theorem 3 follows. $\square$

In general, Theorem 3 cannot be improved, as we see in the next theorem.

Theorem 4 (Johnson [18], van Wee [29]). *For $m \geq 1$ we have*

$$K(2^m, 1) = 2^{2^m - m}.$$

Proof. By Theorem 1 there exists a perfect code $C \subset \mathbb{F}^{2^m - 1}$ of length $2^m - 1$, covering radius one and cardinality $|C| = 2^{2^m - m - 1}$. One easily sees, that the code $C' \subset \mathbb{F}^{2^m}$ of length 2^m defined by

$$C' = \{(0, \mathbf{x}) \mid \mathbf{x} \in C\} \cup \{(1, \mathbf{x}) \mid \mathbf{x} \in C\}$$

has covering radius one and satisfies $|C'| = 2^{2^m - m}$. Therefore $K(2^m, 1) \leq 2^{2^m - m}$. The corresponding lower bound follows from Theorem 3. $\square$

If n is odd, there is no forced covering excess in a sphere of radius one. For instance the Hamming sphere $B(\mathbf{0}, 1)$ centered on the all zero word $\mathbf{0}$ of length n could be perfectly covered (i.e. without covering excess) by the words $\mathbf{e}_1, \mathbf{e}_2 + \mathbf{e}_3, \mathbf{e}_4 + \mathbf{e}_5, \dots, \mathbf{e}_{n-1} + \mathbf{e}_n$. What can we do to improve on the sphere-covering bound? We have to look for a covering excess in a sphere with radius greater than one!

Chapter 3

Bounds for $K(n, 1)$ with odd n

In this chapter we study bounds for $K(n, 1)$, where n is odd. We start with the case $n \equiv 5 \pmod{6}$, where a covering excess in a sphere of radius two can be found. We describe a classical bound of van Wee [29] for this case as well as an improvement due to Blass and Litsyn [1]. We further improve on this bound, giving the presently best known lower bound for $K(n, 1)$ if $n \equiv 5 \pmod{6}$.

Next we discuss the general excess bound for $K(n, 1)$ (due to Habsieger [9] and Honkala [16]), which follows from the fact, that if $n \equiv -1 \pmod{p}$ for an odd prime p , there exists a covering excess $\geq p - 1$ on a sphere with radius $p - 1$. Answering a question of Habsieger, we prove that this covering excess indeed grows quadratic in p . This yields a "general improvement of the general excess bound" for binary codes with covering radius one. The proof uses a classification theorem for certain subset systems as well as new congruence properties for the excess function, which were conjectured by Habsieger.

3.1 The case $n \equiv 5 \pmod{6}$

3.1.1 The bound of van Wee

Let $C \subset \mathbb{F}^n$ be a binary code with covering radius one. As we have already seen, there is no need for a covering excess in a sphere of radius one in case of odd n , so let's hope $n \equiv -1 \pmod{3}$, when the following theorem of van Wee applies.

Theorem 5 (van Wee [29]). *Let $C \subset \mathbb{F}^n$ be a binary code with covering radius one and $n \equiv -1 \pmod{3}$. If $\mathbf{x} \in \mathbb{F}^n$, then*

$$E(B(\mathbf{x}, 2)) \equiv -1 \pmod{3}. \quad (3.1)$$

Proof. If $\mathbf{c} \in C$, then $|B(\mathbf{c}, 1) \cap B(\mathbf{x}, 2)|$ equals the number of words in the sphere $B(\mathbf{x}, 2)$, which were 1-covered by $\mathbf{c}$. One easily sees, that

$$|B(\mathbf{c}, 1) \cap B(\mathbf{x}, 2)| = \begin{cases} |B(\mathbf{c}, 1)| = n + 1 & \text{if } d(\mathbf{c}, \mathbf{x}) \leq 1, \\ 3 & \text{if } d(\mathbf{c}, \mathbf{x}) = 2 \text{ or } 3, \\ 0 & \text{if } d(\mathbf{c}, \mathbf{x}) \geq 4. \end{cases}$$

Now (2.6) applied to $V = B(\mathbf{x}, 2)$ yields

$$\begin{aligned} E(B(\mathbf{x}, 2)) &= \sum_{\mathbf{c} \in C} |B(\mathbf{c}, 1) \cap B(\mathbf{x}, 2)| - |B(\mathbf{x}, 2)| \\ &= (n + 1) \sum_{\mathbf{c} \in C, d(\mathbf{c}, \mathbf{x}) \leq 1} 1 + 3 \sum_{\mathbf{c} \in C, 2 \leq d(\mathbf{c}, \mathbf{x}) \leq 3} 1 - \left(1 + n + \binom{n}{2}\right) \\ &\equiv -\binom{n}{2} \pmod{3} \\ &\equiv -1 \pmod{3} \end{aligned}$$

by (2.1) and $n \equiv -1 \pmod{3}$. □

If $n \equiv -1 \pmod{3}$ with even n , then the better bound in Theorem 3 applies, so we may restrict us to the case $n \equiv 5 \pmod{6}$.

Theorem 6 (van Wee [29]). *When $n \equiv 5 \pmod{6}$, we have*

$$K(n, 1) \geq \left(1 + \frac{4}{n^2 + n + 2}\right) \frac{2^n}{n + 1}.$$

Proof. Let $C \subset \mathbb{F}^n$ be a binary code with covering radius one and $|C| = K(n, 1)$. Since $E(B(\mathbf{x}, 2))$ is always nonnegative for $\mathbf{x} \in \mathbb{F}^n$, from (3.1) follows

$$E(B(\mathbf{x}, 2)) \geq 2 \quad \text{for each } \mathbf{x} \in \mathbb{F}^n. \quad (3.2)$$

We now get

$$\begin{aligned}
2 \cdot 2^n &= \sum_{\mathbf{x} \in \mathbb{F}^n} 2 \\
&\leq \sum_{\mathbf{x} \in \mathbb{F}^n} E(B(\mathbf{x}, 2)) \quad \text{by (3.2)} \\
&= \sum_{\mathbf{x} \in \mathbb{F}^n} \sum_{\mathbf{z} \in B(\mathbf{x}, 2)} E(\mathbf{z}) \\
&= \sum_{\mathbf{z} \in \mathbb{F}^n} E(\mathbf{z}) \sum_{\mathbf{x} \in B(\mathbf{z}, 2)} 1 \\
&= V(n, 2) E(\mathbb{F}^n) \\
&= V(n, 2) (|C|(n+1) - 2^n) \quad \text{by (2.8)}
\end{aligned}$$

and Theorem 6 follows by $V(n, 2) = 1 + n + \binom{n}{2}$ (see (2.1)). $\square$

3.1.2 The Theorem of Blass and Litsyn

Theorem 6 was further improved by Blass and Litsyn [1], Habsieger [9], Honkala [15], [16], [17] and van Wee [31]. The best of these bounds is due to Blass and Litsyn [1]:

Theorem 7 (Blass and Litsyn [1]). *If $n \equiv 5 \pmod{6}$, then*

$$K(n, 1) \geq \left(1 + \frac{40}{10n^2 - 44n + 77}\right) \frac{2^n}{n+1}. \quad (3.3)$$

In section 3.1.3 we give a simplification of the proof of Theorem 7, which at the same time leads to a better result. We give the complete proof (for $n \geq 17$), since it is divided on the three papers [1], [7], [16] and it saves the efforts of the reader to find out, which of the material of [1] is needed and which not. For the rest of this section we give the necessary lemmas, which already can be found in [1] and [16], in a slightly modified manner.

All improvements on Theorem 6 mentioned above rely on the fact, that not always equality can hold in (3.2) and so one has to upper-bound $|B|$, where B is defined by

$$B = \{\mathbf{x} \in \mathbb{F}^n \mid E(B(\mathbf{x}, 2)) = 2\}. \quad (3.4)$$

Crucial is a structural property of B described in the next lemma. Assume for the rest of this section, that $C \subset \mathbb{F}^n$ is a fixed binary code with covering radius one and $n \equiv 5 \pmod{6}$, $n \geq 11$.

Lemma 8 (van Wee [31]). *For every $\mathbf{x} \in B$ there exists a unique $\mathbf{z} \in B(\mathbf{x}, 2)$ with $E(\mathbf{z}) > 0$. Consequently $E(\mathbf{z}) = 2$ (i.e. $\mathbf{z} \in Z_2$).*

Proof. Assume to the contrary, there exist distinct $\mathbf{z}_1, \mathbf{z}_2 \in B(\mathbf{x}, 2)$ with $E(\mathbf{z}_1) > 0$ and $E(\mathbf{z}_2) > 0$. One easily sees, that in all cases there exists $\mathbf{y} \in B(\mathbf{x}, 1)$ such that, say $\mathbf{z}_1 \in B(\mathbf{y}, 1)$, but $\mathbf{z}_2 \notin B(\mathbf{y}, 1)$. Now $E(B(\mathbf{y}, 1)) \geq E(\mathbf{z}_1) \geq 1$. On the other hand by Theorem 2 we have $E(B(\mathbf{y}, 1)) \equiv 0 \pmod{2}$, implying $E(B(\mathbf{y}, 1)) \geq 2$. This however gives the contradiction $2 = E(B(\mathbf{x}, 2)) \geq E(B(\mathbf{y}, 1)) + E(\mathbf{z}_2) \geq 3$. $\square$

For $\mathbf{x} \in \mathbb{F}^n$ we set

$$\begin{aligned} L &= \{\mathbf{y} \in \mathbb{F}^n : E(B(\mathbf{y}, 2)) = 5\}, \\ B(\mathbf{x}) &= B(\mathbf{x}, 2) \cap B. \end{aligned}$$

One easily checks, that if $\mathbf{z} \in Z_2$, then $E(B(\mathbf{z}, 2)) > 2$ and thus $E(B(\mathbf{z}, 2)) \geq 5$ by Theorem 5. We partition Z_2 into the following subsets (different from [1]):

$$\begin{aligned} Z_2^{(1)} &= L \cap Z_2, \\ Z_2^{(2)} &= \left\{ \mathbf{z} \in Z_2 \setminus Z_2^{(1)} \mid |B(\mathbf{z})| > \binom{n-3}{2} \right\}, \\ Z_2^{(3)} &= \left\{ \mathbf{z} \in Z_2 \setminus Z_2^{(1)} \mid |B(\mathbf{z})| \leq \binom{n-3}{2} \right\}. \end{aligned} \quad (3.5)$$

The next lemmas are from [1].

Lemma 9 (Blass, Litsyn [1]). *If $\mathbf{z} \in Z_2$, then*

$$|B(\mathbf{z})| \leq \binom{n-2}{2}. \quad (3.6)$$

If additionally $\mathbf{z} \in Z_2^{(1)}$, then there are exactly three distinct words $\mathbf{y}_i \in B(\mathbf{z}, 2)$, $i = 1, 2, 3$, such that $E(\mathbf{y}_i) = 1$. For every $\mathbf{c} \in C$ with $d(\mathbf{y}_i, \mathbf{c}) \leq 1$ for at least one $\mathbf{y}_i$ we also have $d(\mathbf{z}, \mathbf{c}) \leq 1$.

Proof. By $\mathbf{z} \in Z_2$, $\mathbf{z}$ is covered exactly three times.

Case I. $\mathbf{z} \in C$.

Without loss $\mathbf{z} = \mathbf{0}$ and $C' = \{\mathbf{0}, \mathbf{e}_1, \mathbf{e}_2\} \subset C$. Clearly

$$E(\mathbf{e}_1) \geq 1, E(\mathbf{e}_2) \geq 1, E(\mathbf{e}_1 + \mathbf{e}_2) \geq 1. \quad (3.7)$$

To show (3.6), simply note that for $\mathbf{x} \in B(\mathbf{0})$ we have $d(\mathbf{x}, \mathbf{e}_1) > 2$, since otherwise $E(B(\mathbf{x}, 2)) \geq E(\mathbf{0}) + E(\mathbf{e}_1) \geq 3$ contradicting $E(B(\mathbf{x}, 2)) = 2$ for $\mathbf{x} \in B(\mathbf{0}) \subset B$. Similar $d(\mathbf{x}, \mathbf{e}_2) > 2$. Thus every word in $B(\mathbf{0})$ has weight two and starts with two zeros, i.e.

$$B(\mathbf{0}) \subset \{\mathbf{e}_i + \mathbf{e}_j \mid 3 \leq i < j \leq n\}, \quad (3.8)$$

and (3.6) follows in case I.

If $\mathbf{z} = \mathbf{0} \in Z_2^{(1)}$, then $E(B(\mathbf{0}, 2)) = 5$ and by (3.7) in fact we have

$$E(\mathbf{e}_1) = E(\mathbf{e}_2) = E(\mathbf{e}_1 + \mathbf{e}_2) = 1. \quad (3.9)$$

Thus we may set $\mathbf{y}_1 = \mathbf{e}_1$, $\mathbf{y}_2 = \mathbf{e}_2$ and $\mathbf{y}_3 = \mathbf{e}_1 + \mathbf{e}_2$. Assume $\mathbf{c} \in C$ with $d(\mathbf{y}_i, \mathbf{c}) \leq 1$ for at least one $\mathbf{y}_i$. By (3.9), $\mathbf{y}_i$ is covered exactly twice. Since each $\mathbf{y}_i$, $i = 1, 2, 3$, is covered by two words from C' , we find $\mathbf{c} \in C'$ and thus $d(\mathbf{z}, \mathbf{c}) = d(\mathbf{0}, \mathbf{c}) \leq 1$.

Case II. $\mathbf{z} \notin C$.

Without loss $\mathbf{z} = \mathbf{0}$ and $\{\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3\} \subset C$. Clearly $E(\mathbf{e}_1 + \mathbf{e}_2) \geq 1$, $E(\mathbf{e}_1 + \mathbf{e}_3) \geq 1$ and like in case I one shows $d(\mathbf{x}, \mathbf{e}_1 + \mathbf{e}_2) > 2$ and $d(\mathbf{x}, \mathbf{e}_1 + \mathbf{e}_3) > 2$ whenever $\mathbf{x} \in B(\mathbf{0})$. Thus every word in $B(\mathbf{0})$ has weight one or two and starts with three zeros, i.e.

$$B(\mathbf{0}) \subset \{\mathbf{e}_i \mid 4 \leq i \leq n\} \cup \{\mathbf{e}_i + \mathbf{e}_j \mid 4 \leq i < j \leq n\}, \quad (3.10)$$

implying $|B(\mathbf{0})| \leq n - 3 + \binom{n-3}{2} = \binom{n-2}{2}$ and (3.6) follows in case II too.

Setting $\mathbf{y}_1 = \mathbf{e}_1 + \mathbf{e}_2$, $\mathbf{y}_2 = \mathbf{e}_2 + \mathbf{e}_3$ and $\mathbf{y}_3 = \mathbf{e}_1 + \mathbf{e}_3$, the proof of the second half of the Lemma is very similar like in case I. $\square$

Lemma 10 (Blass, Litsyn [1]). *We have*

$$|Z_1| \geq 3|Z_2^{(1)}|. \quad (3.11)$$

Proof. By Lemma 9, $|Z_1 \cap B(\mathbf{z}, 2)| = 3$ whenever $\mathbf{z} \in Z_2^{(1)}$. Thus Lemma 10 follows, if we can show, that no $\mathbf{y} \in Z_1$ satisfies $\mathbf{y} \in B(\mathbf{z}_1, 2) \cap B(\mathbf{z}_2, 2)$ whenever $\mathbf{z}_1, \mathbf{z}_2 \in Z_2^{(1)}$ with $\mathbf{z}_1 \neq \mathbf{z}_2$.

Assume to the contrary, that $\mathbf{y} \in Z_1 \cap B(\mathbf{z}_1, 2) \cap B(\mathbf{z}_2, 2)$. Since $\mathbf{y}$ is covered by C , there exists $\mathbf{c} \in C$ with $d(\mathbf{c}, \mathbf{y}) \leq 1$. By Lemma 9 we get $d(\mathbf{c}, \mathbf{z}_1) \leq 1$ as well as $d(\mathbf{c}, \mathbf{z}_2) \leq 1$, implying $d(\mathbf{z}_1, \mathbf{z}_2) \leq 2$. But then $B(\mathbf{z}_1, 2)$ contains two words from Z_2 ($\mathbf{z}_1$ and $\mathbf{z}_2$) as well as three words from Z_1 by Lemma 9 and thus $E(B(\mathbf{z}_1, 2)) \geq 7$ contradicting $\mathbf{z}_1 \in Z_2^{(1)}$. $\square$

Lemma 11 (Blass, Litsyn [1]). *If $\mathbf{z} \in Z_2^{(2)}$, there are exactly three words $\mathbf{y}_i \in B(\mathbf{z}, 2) \setminus \{\mathbf{z}\}$ ($i = 1, 2, 3$) satisfying $E(\mathbf{y}_i) = 2$.*

Proof. Case I. $\mathbf{z} \in C$.

Without loss $\mathbf{z} = \mathbf{0}$ and $\{\mathbf{0}, \mathbf{e}_1, \mathbf{e}_2\} \subset C$. We set $\mathbf{y}_1 = \mathbf{e}_1$, $\mathbf{y}_2 = \mathbf{e}_2$ and $\mathbf{y}_3 = \mathbf{e}_1 + \mathbf{e}_2$. By (3.7) we have $\mathbf{y}_i \in B(\mathbf{z}, 2) \setminus \{\mathbf{z}\}$ and $E(\mathbf{y}_i) > 0$ ($i = 1, 2, 3$). We next prove, that there is no other word $\mathbf{y} \in B(\mathbf{0}, 2) \setminus \{\mathbf{0}\}$ satisfying $E(\mathbf{y}) > 0$.

Without loss it suffices to consider the cases $\mathbf{y} = \mathbf{e}_3$ or $\mathbf{y} = \mathbf{e}_i + \mathbf{e}_3$ with $i \neq 3$. By $E(\mathbf{e}_3) > 0$ (resp. $E(\mathbf{e}_i + \mathbf{e}_3) > 0$) and $E(\mathbf{0}) = 2$ we have $E(B(\mathbf{e}_3 + \mathbf{e}_j, 2)) > 2$ for $j \neq 3$, implying $\mathbf{e}_3 + \mathbf{e}_j \notin B(\mathbf{0})$. Therefore by (3.8) each word of $B(\mathbf{0})$ must be of the form $\mathbf{e}_i + \mathbf{e}_j$, $4 \leq i < j \leq n$. The number of such vectors is at most $\binom{n-3}{2}$, contradicting the definition of $Z_2^{(2)}$.

It remains to show $E(\mathbf{y}_1) = E(\mathbf{y}_2) = E(\mathbf{y}_3) = 2$. The case $E(\mathbf{y}_1) = E(\mathbf{y}_2) = E(\mathbf{y}_3) = 1$ is impossible, since it yields $E(B(\mathbf{z}, 2)) = 5$ and thus $\mathbf{z} \in Z_2^{(1)}$, contradicting the definition of $Z_2^{(2)}$. If $E(\mathbf{y}_1) \geq 2$ or $E(\mathbf{y}_3) \geq 2$, then $\mathbf{e}_1 + \mathbf{e}$ (with $\mathbf{e} = \mathbf{0}$ or $\mathbf{e} = \mathbf{e}_2$) must be covered by a codeword $\mathbf{c} \in C$, $C \neq \mathbf{0}, \mathbf{e}_1, \mathbf{e}_2$. Now $\mathbf{c} = \mathbf{e}_1 + \mathbf{e} + \mathbf{e}_j$ with $j \geq 3$ is impossible, since then $E(\mathbf{y}) > 0$ with $\mathbf{y} = \mathbf{e}_1 + \mathbf{e}_j$ and $\mathbf{y} \in B(\mathbf{z}, 2) \setminus \{\mathbf{z}, \mathbf{y}_1, \mathbf{y}_2, \mathbf{y}_3\}$. There remains the possibility $\mathbf{c} = \mathbf{e}_1 + \mathbf{e}_2 \in C$ and $E(\mathbf{y}_1) = E(\mathbf{y}_3) = 2$ follows. For reasons of symmetry we also have $E(\mathbf{y}_2) = 2$.

Case II. $\mathbf{z} \notin C$.

Without loss $\mathbf{z} = \mathbf{0}$ and $\{\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3\} \subset C$. We set $\mathbf{y}_1 = \mathbf{e}_1 + \mathbf{e}_2$, $\mathbf{y}_2 = \mathbf{e}_2 + \mathbf{e}_3$ and $\mathbf{y}_3 = \mathbf{e}_1 + \mathbf{e}_3$. Clearly $E(\mathbf{y}_i) > 0$ ($i = 1, 2, 3$). We show, that there is no other word $\mathbf{y} \in B(\mathbf{0}, 2) \setminus \{\mathbf{0}\}$ with $E(\mathbf{y}) > 0$. It suffices to consider the cases $\mathbf{y} = \mathbf{e}_{i_0}$ or $\mathbf{y} = \mathbf{e}_{i_0} + \mathbf{e}_4$, $1 \leq i_0 \leq n$. Similar like in case I we see, that we then have $B(\mathbf{0}) \subset B_1 := \{\mathbf{e}_i + \mathbf{e}_j \mid 4 \leq i < j \leq n\}$ in the first case and $B(\mathbf{0}) \subset B_2 := \{\mathbf{e}_i \mid 5 \leq i \leq n\} \cup \{\mathbf{e}_i + \mathbf{e}_j \mid 5 \leq i < j \leq n\}$ in the second case (use (3.10)). Since $|B_1| = |B_2| = \binom{n-3}{2}$ we get a contradiction to the definition of $Z_2^{(2)}$.

It remains to show $E(\mathbf{y}_1) = E(\mathbf{y}_2) = E(\mathbf{y}_3) = 2$. Like in case I $E(\mathbf{y}_1) = E(\mathbf{y}_2) = E(\mathbf{y}_3) = 1$ is impossible. For reasons of symmetry it suffices to assume $E(\mathbf{y}_1) \geq 2$. Then $\mathbf{y}_1 = \mathbf{e}_1 + \mathbf{e}_2$ must be covered by a codeword $\mathbf{c} \in C$, $\mathbf{c} = \mathbf{e}_1 + \mathbf{e}_2 + \mathbf{e}$ with $\mathbf{e} = \mathbf{0}$ or $\mathbf{e} = \mathbf{e}_i$, $i \geq 3$. Now $\mathbf{e} = \mathbf{0}$ (resp. $\mathbf{e} = \mathbf{e}_i$, $i \geq 4$) is impossible, since then $E(\mathbf{e}_1) > 0$ (resp. $E(\mathbf{e}_1 + \mathbf{e}_i) > 0$). Hence $\mathbf{y}_1 \notin C$, $\mathbf{c} = \mathbf{e}_1 + \mathbf{e}_2 + \mathbf{e}_3 \in C$ and $E(\mathbf{y}_1) = 2$ follows. Since $\mathbf{c}$ also covers $\mathbf{y}_2$ and $\mathbf{y}_3$, we get $E(\mathbf{y}_2) \geq 2$ and $E(\mathbf{y}_3) \geq 2$. Now like in the case of $\mathbf{y}_1$ we have

$E(\mathbf{y}_2) = E(\mathbf{y}_3) = 2$ also. □

For $\mathbf{z} \in Z_2^{(2)}$ we set

$$H(\mathbf{z}) = \{\mathbf{z}, \mathbf{y}_1, \mathbf{y}_2, \mathbf{y}_3\} \subset Z_2^{(2)} \cup Z_2^{(3)}, \quad (3.12)$$

where $\mathbf{y}_1, \mathbf{y}_2$ and $\mathbf{y}_3$ are defined in Lemma 11. Clearly

$$|H(\mathbf{z})| = 4 \quad \text{if } \mathbf{z} \in Z_2^{(2)}. \quad (3.13)$$

Following the proof of Lemma 11, we notice, that if $\mathbf{z} \in C$, then for some $i, j \in \{1, \dots, n\}$, $i \neq j$

$$H(\mathbf{z}) = \{\mathbf{z}, \mathbf{z} + \mathbf{e}_i, \mathbf{z} + \mathbf{e}_j, \mathbf{z} + \mathbf{e}_i + \mathbf{e}_j\} \subset C. \quad (3.14)$$

If $\mathbf{z} \notin C$, then

$$H(\mathbf{z}) = \{\mathbf{z}, \mathbf{z} + \mathbf{e}_k + \mathbf{e}_l, \mathbf{z} + \mathbf{e}_k + \mathbf{e}_m, \mathbf{z} + \mathbf{e}_l + \mathbf{e}_m\} \text{ with } H(\mathbf{z}) \cap C = \emptyset \quad (3.15)$$

for some $k, l, m \in \{1, \dots, n\}$, $k \neq l \neq m \neq k$. Moreover in that case $\{\mathbf{z} + \mathbf{e}_k, \mathbf{z} + \mathbf{e}_l, \mathbf{z} + \mathbf{e}_m, \mathbf{z} + \mathbf{e}_k + \mathbf{e}_l + \mathbf{e}_m\} \subset C$.

Lemma 12 (Blass, Litsyn [1]). *If $\mathbf{z}_1, \mathbf{z}_2 \in Z_2^{(2)}$, then either $H(\mathbf{z}_1) \cap H(\mathbf{z}_2) = \emptyset$ or $H(\mathbf{z}_1) = H(\mathbf{z}_2)$.*

Proof. Assume $\mathbf{y} \in H(\mathbf{z}_1) \cap H(\mathbf{z}_2)$. By definition of $H(\mathbf{z})$ we have $E(\mathbf{y}) = 2$.

Case I. $\mathbf{y} \in C$.

Then $\mathbf{z}_1, \mathbf{z}_2 \in C$ (otherwise $H(\mathbf{z}_1) \cap C = \emptyset$). From (3.14) follows $H(\mathbf{z}_1) = \{\mathbf{z}_1, \mathbf{z}_1 + \mathbf{e}_i, \mathbf{z}_1 + \mathbf{e}_j, \mathbf{z}_1 + \mathbf{e}_i + \mathbf{e}_j\} = \{\mathbf{y}, \mathbf{y} + \mathbf{e}_i, \mathbf{y} + \mathbf{e}_j, \mathbf{y} + \mathbf{e}_i + \mathbf{e}_j\} \subset C$ for some $i, j \in \{1, \dots, n\}$, $i \neq j$. Similar by $\mathbf{y} \in H(\mathbf{z}_2)$ we have $H(\mathbf{z}_2) = \{\mathbf{y}, \mathbf{y} + \mathbf{e}_k, \mathbf{y} + \mathbf{e}_l, \mathbf{y} + \mathbf{e}_k + \mathbf{e}_l\} \subset C$ for some $k, l \in \{1, \dots, n\}$, $k \neq l$. We now get $\{i, j\} = \{k, l\}$ (otherwise $E(\mathbf{y}) > 2$), i.e. $H(\mathbf{z}_1) = H(\mathbf{z}_2)$.

Case II. $\mathbf{y} \notin C$.

Then $\mathbf{z}_1, \mathbf{z}_2 \notin C$ (otherwise $H(\mathbf{z}_1), H(\mathbf{z}_2) \subset C$). From (3.15) follows $H(\mathbf{z}_1) = \{\mathbf{z}_1, \mathbf{z}_1 + \mathbf{e}_k + \mathbf{e}_l, \mathbf{z}_1 + \mathbf{e}_k + \mathbf{e}_m, \mathbf{z}_1 + \mathbf{e}_l + \mathbf{e}_m\} = \{\mathbf{y}, \mathbf{y} + \mathbf{e}_k + \mathbf{e}_l, \mathbf{y} + \mathbf{e}_k + \mathbf{e}_m, \mathbf{y} + \mathbf{e}_l + \mathbf{e}_m\}$ for some $k, l, m \in \{1, \dots, n\}$, $k \neq l \neq m \neq k$ and $\{\mathbf{z}_1 + \mathbf{e}_k, \mathbf{z}_1 + \mathbf{e}_l, \mathbf{z}_1 + \mathbf{e}_m, \mathbf{z}_1 + \mathbf{e}_k + \mathbf{e}_l + \mathbf{e}_m\} = \{\mathbf{y} + \mathbf{e}_k, \mathbf{y} + \mathbf{e}_l, \mathbf{y} + \mathbf{e}_m, \mathbf{y} + \mathbf{e}_k + \mathbf{e}_l + \mathbf{e}_m\} \subset C$. Similar by $\mathbf{y} \in H(\mathbf{z}_2)$ we have $H(\mathbf{z}_2) = \{\mathbf{y}, \mathbf{y} + \mathbf{e}_{k'} + \mathbf{e}_{l'}, \mathbf{y} + \mathbf{e}_{k'} + \mathbf{e}_{m'}, \mathbf{y} + \mathbf{e}_{l'} + \mathbf{e}_{m'}\}$ for some $k', l', m' \in \{1, \dots, n\}$, $k' \neq l' \neq m' \neq k'$ and $\{\mathbf{y} + \mathbf{e}_{k'}, \mathbf{y} + \mathbf{e}_{l'}, \mathbf{y} + \mathbf{e}_{m'}, \mathbf{y} + \mathbf{e}_{k'} + \mathbf{e}_{l'} + \mathbf{e}_{m'}\} \subset C$. Like in case I we now get $\{k, l, m\} = \{k', l', m'\}$, i.e. $H(\mathbf{z}_1) = H(\mathbf{z}_2)$. □

The proof of the case $\mathbf{z} \in C$ of the following Lemma already appeared in Honkala [16].

Lemma 13 (Blass, Litsyn [1], Honkala [16]). *If $\mathbf{z} \in Z_2^{(2)}$ we have*

$$\sum_{\mathbf{y} \in H(\mathbf{z})} |B(\mathbf{y})| \leq 4 \binom{n-3}{2}. \quad (3.16)$$

Proof. By the definition of $H(\mathbf{z})$ we have $\mathbf{y} \in Z_2$ whenever $\mathbf{y} \in H = H(\mathbf{z})$. Indeed we will show

$$\sum_{\mathbf{y} \in H \setminus \{\mathbf{x}\}} |B(\mathbf{y})| \leq 3 \binom{n-3}{2} \quad (3.17)$$

whenever $\mathbf{x} \in H$, since by (3.13) this implies

$$3 \sum_{\mathbf{y} \in H} |B(\mathbf{y})| = \sum_{\mathbf{x} \in H} \sum_{\mathbf{y} \in H \setminus \{\mathbf{x}\}} |B(\mathbf{y})| \leq 12 \binom{n-3}{2}$$

and (3.16) follows.

Case I. $\mathbf{z} \in C$.

Without loss $\mathbf{z} = \mathbf{0}$ and $H = \{\mathbf{0}, \mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_1 + \mathbf{e}_2\}$, see (3.14). Moreover, it suffices to show (3.17) for $\mathbf{x} = \mathbf{e}_1 + \mathbf{e}_2$, i.e. $H' := H \setminus \{\mathbf{x}\} = \{\mathbf{0}, \mathbf{e}_1, \mathbf{e}_2\}$. For $\mathbf{y} \in H'$ we have

$$B(\mathbf{y}) \subset B^{(\mathbf{y})} := \{\mathbf{y} + \mathbf{e}_i + \mathbf{e}_j \mid 3 \leq i < j \leq n\}, \quad (3.18)$$

(compare (3.8)). Note, that the sets $B^{(\mathbf{y})}$, $\mathbf{y} \in H'$ are pairwise disjoint.

We now consider the covering excess in a sphere of radius two centered on $\mathbf{e}_i$, $3 \leq i \leq n$. Clearly $B(\mathbf{e}_i, 2) \cap H = H'$ and thus $E(B(\mathbf{e}_i, 2) \cap H) = 6$ by $E(\mathbf{y}) = 2$ whenever $\mathbf{y} \in H$. By Theorem 5 however we have $E(B(\mathbf{e}_i, 2)) \equiv -1 \pmod{3}$. This implies

$$\forall 3 \leq i \leq n \exists \mathbf{u}_i \in B(\mathbf{e}_i, 2) \setminus H \text{ with } E(\mathbf{u}_i) > 0. \quad (3.19)$$

We set $B^* = \bigcup_{\mathbf{y} \in H'} B^{(\mathbf{y})}$ and consider two subcases.

Subcase A. Each word $\mathbf{u}_i$ has weight at least two, $3 \leq i \leq n$.

We first show

$$|B(\mathbf{u}_i, 2) \cap B^* \cap \mathbb{F}_i^n| \geq 6 \quad (3 \leq i \leq n). \quad (3.20)$$

To see this, fix i with $3 \leq i \leq n$. Note that in subcase A by $\mathbf{u}_i \in B(\mathbf{e}_i, 2)$ we have $\mathbf{u}_i \in \mathbb{F}_i^n$. If $\mathbf{u}_i \in \mathbb{F}_1^n$, then $B(\mathbf{u}_i, 2) \cap B^{(\mathbf{e}_1)} \cap \mathbb{F}_i^n = B^{(\mathbf{e}_1)} \cap \mathbb{F}_i^n$ implying $|B(\mathbf{u}_i, 2) \cap B^{(\mathbf{e}_1)} \cap \mathbb{F}_i^n| = |B^{(\mathbf{e}_1)} \cap \mathbb{F}_i^n| = n - 3 \geq 6$ and (3.20) follows. If $\mathbf{u}_i \in \mathbb{F}_2^n$ or $w(\mathbf{u}_i) = 2$, then the argument is very similar. There remains the case, that $\mathbf{u}_i$ starts with two zeros and satisfies $w(\mathbf{u}_i) = 3$. Then for each of the three elements $\mathbf{y} \in H'$ we have $|B(\mathbf{u}_i, 2) \cap B^{(\mathbf{y})} \cap \mathbb{F}_i^n| = 2$ and (3.20) follows again.

Each word in B^* occurs in exactly two of the sets $B^* \cap \mathbb{F}_i^n$, $3 \leq i \leq n$. Therefore

$$\begin{aligned} & \left| \bigcup_{3 \leq i \leq n} (B(\mathbf{u}_i, 2) \cap B^*) \right| \\ & \geq \left| \bigcup_{3 \leq i \leq n} (B(\mathbf{u}_i, 2) \cap B^* \cap \mathbb{F}_i^n) \right| \geq \frac{1}{2} \sum_{3 \leq i \leq n} |B(\mathbf{u}_i, 2) \cap B^* \cap \mathbb{F}_i^n| \geq 3(n - 3). \end{aligned} \quad (3.21)$$

by (3.20). Now, if $\mathbf{x} \in B(\mathbf{u}_i, 2) \cap B^*$, ($3 \leq i \leq n$), then there exists $\mathbf{y}' \in H'$ with $\mathbf{x} \in B^{(\mathbf{y}')}$ and thus $\mathbf{y}' \in B(\mathbf{x}, 2)$. Therefore $E(B(\mathbf{x}, 2)) \geq E(\mathbf{y}') + E(\mathbf{u}_i) > 2$, which implies $\mathbf{x} \notin \bigcup_{\mathbf{y} \in H'} B(\mathbf{y})$ whenever $\mathbf{x} \in B(\mathbf{u}_i, 2) \cap B^*$, $3 \leq i \leq n$. Using (3.18) and (3.21) we now get

$$\begin{aligned} \sum_{\mathbf{y} \in H'} |B(\mathbf{y})| & \leq |B^*| - \left| \bigcup_{3 \leq i \leq n} (B(\mathbf{u}_i, 2) \cap B^*) \right| \\ & \leq 3 \binom{n-2}{2} - 3(n-3) = 3 \binom{n-3}{2} \end{aligned}$$

and (3.17) follows, completing the proof of Lemma 13 in subcase A.

Subcase B. There is a word $\mathbf{u}_i$ with $w(\mathbf{u}_i) = 1$.

If $\mathbf{e}_{j_0}$ with a suitable j_0 , $3 \leq j_0 \leq n$ occurs among the words $\mathbf{u}_i$, then (3.21) follows again, since then each of the $3(n-3)$ words of the form $\mathbf{y} + \mathbf{e}_i + \mathbf{e}_{j_0}$, $\mathbf{y} \in H'$, $3 \leq i \leq n$, $i \neq j_0$ lies in $B(\mathbf{e}_{j_0}, 2) \cap B^*$. Since we have already seen, that (3.21) implies (3.17), the proof of Lemma 13 is complete in case I.

Case II. $\mathbf{z} \notin C$.

In this case by (3.15) we may assume $\mathbf{z} = \mathbf{0}$ and $H = \{\mathbf{0}, \mathbf{e}_1 + \mathbf{e}_2, \mathbf{e}_1 + \mathbf{e}_3, \mathbf{e}_2 + \mathbf{e}_3\}$. It suffices to show (3.17) for $\mathbf{x} = \mathbf{e}_2 + \mathbf{e}_3$, i.e. $H' = H \setminus \{\mathbf{x}\} = \{\mathbf{0}, \mathbf{e}_1 + \mathbf{e}_2, \mathbf{e}_1 + \mathbf{e}_3\}$. For $\mathbf{y} \in H'$ we have

$$B(\mathbf{y}) \subset B^{(\mathbf{y})} := \{\mathbf{y} + \mathbf{e}_i \mid 4 \leq i \leq n\} \cup \{\mathbf{y} + \mathbf{e}_i + \mathbf{e}_j \mid 4 \leq i < j \leq n\},$$

see (3.10). The excess argument from case I leading to (3.19) applies equally well to the spheres $B(\mathbf{e}_1 + \mathbf{e}_i, 2)$, $4 \leq i \leq n$ and yields

$$\forall 4 \leq i \leq n \exists \mathbf{v}_i \in B(\mathbf{e}_1 + \mathbf{e}_i, 2) \setminus H \text{ with } E(\mathbf{v}_i) > 0.$$

Again we set $B^* = \bigcup_{\mathbf{y} \in H'} B^{(\mathbf{y})}$ and consider two subcases.

Subcase A. Neither $\mathbf{e}_1$ nor one of the words $\mathbf{e}_1 + \mathbf{e}_j$ occurs among the words $\mathbf{v}_i$, $4 \leq i, j \leq n$.

We show

$$|B(\mathbf{v}_i, 2) \cap B^* \cap \mathbb{F}_i^n| \geq 6 \quad (4 \leq i \leq n). \quad (3.22)$$

To see this, fix i with $4 \leq i \leq n$. Again in subcase A we have $\mathbf{v}_i \in \mathbb{F}_i^n$. If $\mathbf{v}_i$ does not start with (100), then $|B(\mathbf{v}_i, 2) \cap B^{(\mathbf{y})} \cap \mathbb{F}_i^n| = n - 3 \geq 6$ for a suitable $\mathbf{y} \in H'$ and (3.22) follows. If $\mathbf{v}_i$ starts with (100) then $|B(\mathbf{v}_i, 2) \cap B^{(\mathbf{y})} \cap \mathbb{F}_i^n| = 2$ for every $\mathbf{y} \in H'$ and (3.22) follows again.

Now (3.21) (with $\mathbf{u}_i$ replaced by $\mathbf{v}_i$) follows like in case I. The rest of the proof of Lemma 13 in subcase A is the same.

Subcase B. Either $\mathbf{e}_1$ or one of the words $\mathbf{e}_1 + \mathbf{e}_j$, $4 \leq j \leq n$ occurs among the words $\mathbf{v}_i$, $4 \leq i \leq n$.

If $\mathbf{e}_1$ occurs among the words $\mathbf{v}_i$, then each of the $3(n - 3)$ words of the form $\mathbf{y} + \mathbf{e}_i$, $\mathbf{y} \in H'$, $4 \leq i \leq n$ lies in $B(\mathbf{e}_1, 2) \cap B^*$. If $\mathbf{e}_1 + \mathbf{e}_{j_0}$ with a suitable j_0 , $4 \leq j_0 \leq n$ occurs among the words $\mathbf{v}_i$, then each of the $3(n - 3)$ words of the form either $\mathbf{y} + \mathbf{e}_{j_0}$ or $\mathbf{y} + \mathbf{e}_i + \mathbf{e}_{j_0}$, $\mathbf{y} \in H'$, $4 \leq i \leq n$, $i \neq j_0$ lies in $B(\mathbf{e}_1 + \mathbf{e}_{j_0}, 2) \cap B^*$. In both cases the proof proceeds exactly like in case I. $\square$

3.1.3 The new bound

In this section we further improve on (3.3). The bound given in the next theorem appears to be the presently best lower bound on $K(n, 1)$ when $n \equiv 5 \pmod{6}$.

Theorem 14 (Haas [7]). *If $n \geq 11$ and $n \equiv 5 \pmod{6}$, then*

$$K(n, 1) \geq \left(1 + \frac{8}{2n^2 - 10n + 19}\right) \frac{2^n}{n + 1}.$$

As a corollary we get the following new bounds on $K(n, 1)$. The old records in parenthesis follow from (3.3).

Corollary 15.

$$\begin{aligned} K(17, 1) &\geq 7419 \quad (7414 [1]), \\ K(23, 1) &\geq 352827 \quad (352736 [1]), \\ K(29, 1) &\geq 17997161 \quad (17994966 [1]). \end{aligned}$$

Proof of Theorem 14. We use the notations of the previous section. Let n be an integer with $n \geq 11$ and $n \equiv 5 \pmod{6}$. By (3.3) Theorem 14 does hold for $n = 11$, so we may assume $n \geq 17$. Let $C \subset \mathbb{F}^n$ be a binary code of length n with covering radius one.

By $M \subset Z_2^{(2)}$ we denote a set of maximal cardinality satisfying $H(\mathbf{z}_1) \cap H(\mathbf{z}_2) = \emptyset$ whenever $\mathbf{z}_1, \mathbf{z}_2 \in M$ with $\mathbf{z}_1 \neq \mathbf{z}_2$. Moreover we set $H = \bigcup_{\mathbf{z} \in M} H(\mathbf{z})$. Then by Lemma 12, (3.12), (3.13) we have $|H| = 4|M|$ and

$$Z_2^{(2)} \cup Z_2^{(3)} = H \cup (Z_2^{(3)} \setminus H). \quad (3.23)$$

Lemma 16. *If $\mathbf{z} \in L$, then $|B(\mathbf{z}, 2) \cap Z_1| \geq 2$.*

Proof. By $|E(B(\mathbf{z}, 2))| = 5$ there exists $\mathbf{u} \in B(\mathbf{z}, 2)$ with $E(\mathbf{u})$ odd. Fix any $\mathbf{x} \in B(\mathbf{u}, 1) \cap B(\mathbf{z}, 1)$. By Theorem 2 we have $E(B(\mathbf{x}, 1)) \equiv 0 \pmod{2}$. Thus there exists $\mathbf{v} \in B(\mathbf{x}, 1) \subset B(\mathbf{z}, 2)$ with $\mathbf{v} \neq \mathbf{u}$ and $E(\mathbf{v})$ odd. Now, since $E(B(\mathbf{z}, 2))$ is odd, there must exist $\mathbf{w} \in B(\mathbf{z}, 2)$ with $E(\mathbf{w})$ odd, $\mathbf{w} \neq \mathbf{u}$ and $\mathbf{w} \neq \mathbf{v}$. Since $E(\mathbf{u})$, $E(\mathbf{v})$ and $E(\mathbf{w})$ are odd, nonnegative integers adding to at most 5, at least two of them must equal one and Lemma 16 follows. $\square$

We now may complete the proof of Theorem 14. From Lemma 8 follows $B \subset \bigcup_{\mathbf{z} \in Z_2} B(\mathbf{z})$ (remember, that B is defined in (3.4)). Thus by (3.23) we get

$$B \subset \bigcup_{\mathbf{z} \in Z_2} B(\mathbf{z}) = \bigcup_{\mathbf{z} \in Z_2^{(1)}} B(\mathbf{z}) \cup \bigcup_{\mathbf{z} \in M} \bigcup_{\mathbf{y} \in H(\mathbf{z})} B(\mathbf{y}) \cup \bigcup_{\mathbf{z} \in Z_2^{(3)} \setminus H} B(\mathbf{z}),$$

which by (3.5), (3.23), Lemma 9 and Lemma 13 implies

$$\begin{aligned}
|B| &\leq \sum_{\mathbf{z} \in Z_2^{(1)}} |B(\mathbf{z})| + \sum_{\mathbf{z} \in M} \sum_{\mathbf{y} \in H(\mathbf{z})} |B(\mathbf{y})| + \sum_{\mathbf{z} \in Z_2^{(3)} \setminus H} |B(\mathbf{z})| \\
&\leq \binom{n-2}{2} |Z_2^{(1)}| + 4|M| \binom{n-3}{2} + \binom{n-3}{2} |(Z_2^{(2)} \cup Z_2^{(3)}) \setminus H| \\
&= \binom{n-2}{2} |Z_2^{(1)}| + \binom{n-3}{2} |Z_2^{(2)} \cup Z_2^{(3)}| \\
&= \binom{n-3}{2} |Z_2| + (n-3) |Z_2^{(1)}|.
\end{aligned} \tag{3.24}$$

Moreover $E(\mathbb{F}^n) = \sum_{i \geq 0} i |Z_i| \geq 2|Z_2| + |Z_1|$. Thus

$$2|Z_2| \leq E(\mathbb{F}^n) - |Z_1|. \tag{3.25}$$

By Theorem 5 we have

$$\begin{aligned}
\left(1 + n + \binom{n}{2}\right) E(\mathbb{F}^n) &= V(n, 2) E(\mathbb{F}^n) \\
&= \sum_{\mathbf{x} \in \mathbb{F}^n} E(B(\mathbf{x}, 2)) \\
&\geq 2|B| + 5|L| + 8(2^n - |B| - |L|) \\
&= 8 \cdot 2^n - 6|B| - 3|L|.
\end{aligned}$$

Substituting (3.11), (3.24) and (3.25) on the right-hand side yields

$$\begin{aligned}
&\left(1 + n + \binom{n}{2} + 3 \binom{n-3}{2}\right) E(\mathbb{F}^n) \\
&\geq 8 \cdot 2^n + \left(3 \binom{n-3}{2} - 2(n-3)\right) |Z_1| - 3|L|.
\end{aligned} \tag{3.26}$$

By Lemma 16 for $n \geq 17$ we have

$$\begin{aligned}
\left(3 \binom{n-3}{2} - 2(n-3)\right) |Z_1| &\geq \frac{3}{2} \left(1 + n + \binom{n}{2}\right) |Z_1| \\
&= \frac{3}{2} \sum_{\mathbf{z} \in Z_1} \sum_{\mathbf{x} \in B(\mathbf{z}, 2)} 1 \\
&\geq \frac{3}{2} \sum_{\mathbf{x} \in L} \sum_{\mathbf{z} \in B(\mathbf{x}, 2) \cap Z_1} 1 \\
&\geq 3|L|.
\end{aligned}$$

This inserted in (3.26) yields

$$E(\mathbb{F}^n) \geq \frac{8 \cdot 2^n}{1 + n + \binom{n}{2} + 3 \binom{n-3}{2}}$$

and Theorem 14 follows by (2.8). $\square$

3.2 The case $n \equiv 1, 3 \pmod{6}$

We have seen, that in the case $n \equiv 5 \pmod{6}$ a binary code of length n and covering radius one forces a covering excess in every sphere of radius two (Theorem 5). In this section we have a look on the remaining odd cases $n \equiv 1, 3 \pmod{6}$ and show, that then there are coverings of spheres with radius three indeed without covering excess. For the proof we have to review shortly some basic facts on Steiner systems.

3.2.1 Steiner systems

We shortly give the definition and a few basic facts for Steiner systems, which are basic objects in design theory. They are related to the construction of perfect coverings of Hamming spheres, i.e. covering codes without covering excess on certain spheres.

Definition 17. *A Steiner system $S(t, k, n)$ is a collection of k -subsets (called blocks) of $\{1, \dots, n\}$, such that every t -subset of $\{1, \dots, n\}$ is contained in exactly one of the blocks.*

The connection to coding theory is given by the quantity *supp* defined in (2.3), which yields a natural one to one correspondence between the words $\mathbf{x} \in \mathbb{F}^n$ and the subsets of $\{1, \dots, n\}$.

The history of Steiner systems goes back until 1847, when Kirkman considered the special case $t = 2$ and $k = 3$. A Steiner system $S(2, 3, n)$ is called a Steiner triple system.

Theorem 18 (Kirkman [20] 1847). *A Steiner triple system $S(2, 3, n)$ exists if and only if $n \equiv 1, 3 \pmod{6}$.*

A Steiner system $S(3, 4, n)$ is called a Steiner quadrupel system.

Theorem 19 (Hanani [12] 1960). *A Steiner quadrupel system $S(3, 4, n)$ exists if and only if $n \equiv 2, 4 \pmod{6}$.*

3.2.2 Perfect coverings of spheres

Definition 20. A perfect R -covering of a Hamming sphere $B(\mathbf{x}, r)$ with $\mathbf{x} \in \mathbb{F}^n$ is a code $C \subset \mathbb{F}^n$, such that each $\mathbf{y} \in B(\mathbf{x}, r)$ is R -covered exactly once by C .

Clearly a perfect code $C \subset \mathbb{F}^n$ with covering radius R is a perfect R -covering of any sphere $B(\mathbf{x}, r)$, $\mathbf{x} \in \mathbb{F}^n$, $r \geq 0$. Perfect coverings of spheres are related to Steiner systems as we will show in the next theorem.

Theorem 21. A perfect 1-covering of a Hamming sphere $B \subset \mathbb{F}^n$ with radius three exists, if and only if there exists a Steiner quadruple system $S(3, 4, n + 1)$, i.e. if $n \equiv 1, 3 \pmod{6}$.

Proof. It suffices to consider the case $B = B(\mathbf{0}, 3)$ with the all-zero word $\mathbf{0} \in \mathbb{F}^n$.

Assume $D_1, \dots, D_r \subset \{1, \dots, n + 1\}$ are the blocks of a Steiner quadruple system $S(3, 4, n + 1)$ on the set $\{1, \dots, n + 1\}$. Then one easily sees, that the code $C = \{\mathbf{0}, \text{supp}^{-1}(D_1 \setminus \{n + 1\}), \dots, \text{supp}^{-1}(D_r \setminus \{n + 1\})\} \subset \mathbb{F}^n$ is a perfect 1-covering of $B(\mathbf{0}, 3)$.

Conversely assume that $C \subset \mathbb{F}^n$ is a perfect 1-covering of $B(\mathbf{0}, 3)$. Let $C_i \subset C$ be the words of weight i in C , $i \geq 0$. If $\mathbf{0} \in C$, then $C_1 = C_2 = \emptyset$. One easily sees, that in this case $\{\text{supp}(\mathbf{u}) \cup \{n + 1\} \mid \mathbf{u} \in C_3\} \cup \{\text{supp}(\mathbf{u}) \mid \mathbf{u} \in C_4\}$ is a Steiner quadruple system $S(3, 4, n + 1)$ on the set $\{1, \dots, n + 1\}$.

Consider the case $\mathbf{0} \notin C$. Then $|C_1| = 1$, say $C_1 = \{\mathbf{e}_n\}$. Then $\text{supp}(\mathbf{u}) \subset \{1, \dots, n - 1\}$ and $|\mathbf{u} \cup \mathbf{v}| = 4$ whenever $\mathbf{u}, \mathbf{v} \in C_2$, $\mathbf{u} \neq \mathbf{v}$. One checks, that

$$\begin{aligned} & \{\mathbf{u} \cup \mathbf{v} \mid \mathbf{u}, \mathbf{v} \in C_2, \mathbf{u} \neq \mathbf{v}\} \\ & \cup \{\text{supp}(\mathbf{u}) \cup \{n, n + 1\} \mid \mathbf{u} \in C_2\} \\ & \cup \{\text{supp}(\mathbf{u}) \cup \{n + 1\} \mid \mathbf{u} \in C_3\} \\ & \cup \{\text{supp}(\mathbf{u}) \mid \mathbf{u} \in C_4\} \end{aligned}$$

is a Steiner quadruple system $S(3, 4, n + 1)$ on the set $\{1, \dots, n + 1\}$.

The last part of the claim now follows from Theorem 19. $\square$

Similar (somewhat easier) one can show, that a perfect 1-covering of a Hamming sphere $B \subset \mathbb{F}^n$ with radius two exists if and only if a Steiner triple system $S(2, 3, n)$ exists, which by Theorem 18 is the case, if $n \equiv 1, 3 \pmod{6}$.

One might try to extend the construction from Theorem 21 to spheres of radius four, thereby obtaining the statement, that a perfect 1-covering of

a Hamming sphere $B \subset \mathbb{F}^n$ with radius four exists if and only if a Steiner quintuple system $S(4, 5, n + 2)$ exists. This however in general is incorrect, since for instance by Theorem 1 there exists a perfect 1-covering of a Hamming sphere $B \subset \mathbb{F}^{15}$ with radius four, but there is no Steiner system $S(4, 5, 17)$ (Östergård and Pottonen [27]). At the present nothing seems to be known concerning perfect 1-coverings of spheres with radius greater than three.

Question 22. *For which values of n there exists a perfect 1-covering of a Hamming sphere $B \subset \mathbb{F}^n$ with radius four?*

By Theorem 21 there is no forced covering excess on a sphere with radius three if $n \equiv 1, 3 \pmod{6}$. In order to improve on the sphere-covering bound (2.9) we will have to look on a covering excess in a sphere with radius greater than three!

3.3 The general excess bound

Again let $C \subset \mathbb{F}^n$ be a code with covering radius one.

Theorem 5 was independently generalized by Habsieger [9] and Honkala [16], who showed that there is a covering excess in a sphere of radius $p - 1$ whenever $n \equiv -1 \pmod{p}$ for an odd prime p . For the proof we need a simple lemma, which was first stated in Cohen et al. [5]. For $\mathbf{x} \in \mathbb{F}^n$ and an integer i we define $A_i(\mathbf{x})$ by

$$A_i(\mathbf{x}) = |\{\mathbf{c} \in C \mid d(\mathbf{c}, \mathbf{x}) = i\}|. \quad (3.27)$$

Note that $A_i(\mathbf{x}) = 0$, if i is negative.

Lemma 23 (Cohen et al. [5]). *For $\mathbf{x} \in \mathbb{F}^n$ and $i \geq 0$ we have*

$$E_i(\mathbf{x}) = (n - i + 1)A_{i-1}(\mathbf{x}) + A_i(\mathbf{x}) + (i + 1)A_{i+1}(\mathbf{x}) - \binom{n}{i}. \quad (3.28)$$

Proof. It suffices to consider the case $\mathbf{x} = \mathbf{0}$. Then $A_j(\mathbf{0})$ is the number of codewords of weight j , $j \geq 0$. Let $S(\mathbf{0}, i)$ denote the set of words with weight i . One easily sees, that for $\mathbf{c} \in C$

$$|B(\mathbf{c}, 1) \cap S(\mathbf{0}, i)| = \begin{cases} n - i + 1 & \text{if } w(\mathbf{c}) = i - 1, \\ 1 & \text{if } w(\mathbf{c}) = i, \\ i + 1 & \text{if } w(\mathbf{c}) = i + 1, \\ 0 & \text{otherwise.} \end{cases}$$

Therefore by (2.6) applied to $V = S(\mathbf{0}, i)$ and $R = 1$ we have

$$\begin{aligned}
E_i(\mathbf{0}) &= E(S(\mathbf{0}, i)) \\
&= \sum_{\mathbf{c} \in C} |B(\mathbf{c}, 1) \cap S(\mathbf{0}, i)| - |S(\mathbf{0}, i)| \\
&= \sum_j \sum_{\substack{\mathbf{c} \in C \\ w(\mathbf{c})=j}} |B(\mathbf{c}, 1) \cap S(\mathbf{0}, i)| - |S(\mathbf{0}, i)| \\
&= (n - i + 1)A_{i-1}(\mathbf{0}) + A_i(\mathbf{0}) + (i + 1)A_{i+1}(\mathbf{0}) - \binom{n}{i}.
\end{aligned}$$

□

Theorem 24 (Habsieger [9], Honkala [16]). *Let $C \subset \mathbb{F}^n$ be a binary code with covering radius one and $n \equiv -1 \pmod{p}$ for an odd prime p . If $\mathbf{x} \in \mathbb{F}^n$, then*

$$E(B(\mathbf{x}, p-1)) = E_0(\mathbf{x}) + E_1(\mathbf{x}) + \dots + E_{p-1}(\mathbf{x}) \equiv -1 \pmod{p}. \quad (3.29)$$

Proof. Summing (3.28) with $0 \leq i \leq p-1$ gives

$$\begin{aligned}
&\sum_{0 \leq i \leq p-1} E_i(\mathbf{x}) \\
&= \sum_{0 \leq i \leq p-1} (n - i + 1)A_{i-1}(\mathbf{x}) + \sum_{0 \leq i \leq p-1} A_i(\mathbf{x}) + \sum_{0 \leq i \leq p-1} (i + 1)A_{i+1}(\mathbf{x}) - \\
&\quad - \sum_{0 \leq i \leq p-1} \binom{n}{i} \\
&= (n + 1) \sum_{0 \leq i \leq p-2} A_i(\mathbf{x}) + p(A_{p-1}(\mathbf{x}) + A_p(\mathbf{x})) - \sum_{0 \leq i \leq p-1} \binom{n}{i} \\
&\equiv - \sum_{0 \leq i \leq p-1} \binom{n}{i} \pmod{p} \\
&\equiv - \sum_{0 \leq i \leq p-1} (-1)^i \pmod{p} \\
&\equiv -1 \pmod{p}
\end{aligned}$$

by $n \equiv -1 \pmod{p}$. □

As a consequence we get the following bound on $K(n, 1)$, the general excess bound for binary codes with covering radius one.

Theorem 25 (Habsieger [9], Honkala [16]). *When $n \equiv -1 \pmod{p}$ for an odd prime p , then*

$$K(n, 1) \geq \left(1 + \frac{p-1}{V(n, p-1)}\right) \frac{2^n}{n+1}. \quad (3.30)$$

Proof. Let $C \subset \mathbb{F}^n$ be a binary code with covering radius one and $|C| = K(n, 1)$. From (3.29) follows $E(B(\mathbf{x}, p-1)) \geq p-1$, since $E(B(\mathbf{x}, p-1))$ is always nonnegative for $\mathbf{x} \in \mathbb{F}^n$. We now get

$$\begin{aligned} (p-1)2^n &= \sum_{\mathbf{x} \in \mathbb{F}^n} (p-1) \\ &\leq \sum_{\mathbf{x} \in \mathbb{F}^n} E(B(\mathbf{x}, p-1)) \\ &= \sum_{\mathbf{x} \in \mathbb{F}^n} \sum_{\mathbf{z} \in B(\mathbf{x}, p-1)} E(\mathbf{z}) \\ &= \sum_{\mathbf{z} \in \mathbb{F}^n} E(\mathbf{z}) \sum_{\mathbf{x} \in B(\mathbf{z}, p-1)} 1 \\ &= V(n, p-1) E(\mathbb{F}^n) \\ &= V(n, p-1)(|C|(n+1) - 2^n) \quad \text{by (2.8)} \end{aligned}$$

and Theorem 25 follows. □

The condition $n \equiv -1 \pmod{p}$ is always satisfied for a suitable odd prime p , except if $n = 2^m - 1$, but then the value of $K(n, 1)$ is known by Theorem 1. If we define the residue classes $\mathcal{R}_p \subset \mathbb{N}$ for an odd prime p by

$$\mathcal{R}_p = \{n \in \mathbb{N} \mid n \equiv -1 \pmod{p}\},$$

we have

$$\mathbb{N} \setminus \{2^m - 1 \mid m \in \mathbb{N}\} = \mathcal{R}_3 \cup \mathcal{R}_5 \cup \dots \cup \mathcal{R}_p \cup \dots$$

Then, as a major statement, the general excess bound says

$$K(n, 1) - \frac{2^n}{n+1} \rightarrow \infty \quad \text{if } n \rightarrow \infty, n \in \mathcal{R}_p \text{ for a fixed odd prime } p. \quad (3.31)$$

Note, that (3.31) (and (3.30)) does not imply

$$K(n, 1) - \frac{2^n}{n+1} \rightarrow \infty \quad \text{if } n \rightarrow \infty, n \in \mathbb{N} \setminus \{2^m - 1 \mid m \in \mathbb{N}\}, \quad (3.32)$$

since (3.30) cannot improve on the sphere covering bound, if p is too large compared to n . The quality of the general excess bound depends on the least odd prime divisor of $n+1$.

A proof of (3.32) could be found in [6].

The rest of this chapter is devoted to the possibility for further improvements of the general excess bound.

3.4 Habsiegers improvement and the excess matrix

Let $C \subset \mathbb{F}^n$ be a code with covering radius one, n odd. Theorem 24 states, that whenever $n \equiv -1 \pmod{p}$ for an odd prime p , then there exists a covering excess in a sphere with radius $p-1$. This gives rise to the natural question, under which circumstances there exists a perfect 1-covering of a sphere with radius $p-2$. A necessary condition is, that p is the least odd prime satisfying $n \equiv -1 \pmod{p}$, since if $n \equiv -1 \pmod{p'}$ with an odd prime $p' < p$, then by Theorem 24 there exists a covering excess in a sphere of radius $p'-1 \leq p-2$. This condition indeed is sufficient in the cases $p=3$ (see the last paragraph of section 2.3) and $p=5$ (see Theorem 21). At the present state of knowledge the existence of a perfect 1-covering of a sphere with radius $p-2$, where n is odd and p is the least odd prime divisor of $n+1$ cannot be proved or refuted when $p \geq 7$, see section 3.2.2. It is perhaps worth noting, that another necessary condition for $E_i(\mathbf{x}) = 0$ whenever $0 \leq i \leq p-2$ arising from (3.28) is always satisfied.

Theorem 26. *Let $n \neq 2^m - 1$ be odd and p the least odd prime satisfying $n \equiv -1 \pmod{p}$. Then there exist nonnegative integers $A_0, \dots, A_{p-1}$ satisfying*

$$E_i := (n - i + 1)A_{i-1} + A_i + (i + 1)A_{i+1} - \binom{n}{i} = 0 \quad (3.33)$$

for $0 \leq i \leq p-2$ (assume $A_{-1} = 0$).

The proof of Theorem 26 is postponed to section 3.5, where the necessary tools are developed.

If it cannot be excluded, that there is no covering excess in a sphere of radius $p - 2$, what can be done to further improve on the general excess bound?

Let us note the following immediate consequence of Theorem 24.

Theorem 27 (Habsieger [9], Honkala [16]). *Assume $C \subset \mathbb{F}^n$ is a binary code of length n and covering radius one with $n \equiv -1 \pmod{p}$ for an odd prime p . Then for every $\mathbf{x} \in \mathbb{F}^n$*

$$E_0(\mathbf{x}) = E_1(\mathbf{x}) = \dots = E_{p-2}(\mathbf{x}) = 0 \quad (3.34)$$

implies

$$E_{p-1}(\mathbf{x}) \geq p - 1. \quad (3.35)$$

Habsiegers idea was, that if (3.34) cannot be excluded, it could be possible to improve on (3.35). His result is conditional and uses another set of congruence properties of the excess function.

Definition 28. *Assume $C \subset \mathbb{F}^n$ is a binary code of length n and covering radius one with odd n . We say, that C satisfies condition (q), if there exist integers $\beta_0, \dots, \beta_{q-2}$, such that*

$$\sum_{0 \leq i \leq q-2} \beta_i E_i(\mathbf{x}) + E_{q-1}(\mathbf{x}) \equiv 0 \pmod{q} \quad (3.36)$$

holds whenever $\mathbf{x} \in \mathbb{F}^n$.

Theorem 29 (Habsieger [9]). *If $p \geq 5$, then in Theorem 27, (3.35) may be replaced by*

$$E_{p-1}(\mathbf{x}) \geq 2p - 1, \quad (3.37)$$

if additionally C satisfies condition (q) for $q = p - 1$ and $q = p - 2$.

We give the proof of Theorem 29 in a framework different from the original, since we develop material needed later. We first introduce the notion of the excess matrix. For the study of covering excess it suffices to consider spheres centered on $\mathbf{0}$ and we use the abbreviation $E_i = E_i(\mathbf{0})$ for $i \geq 0$.

Definition 30. Let $C \subset \mathbb{F}^n$ be a binary code of length n and covering radius one with $n \equiv -1 \pmod{p}$ for an odd prime p . Assume

$$E_0 = E_1 = \dots = E_{p-2} = 0. \quad (3.38)$$

Let $\mathbf{d}_j$ ($j \in A = \{1, \dots, E_{p-1}\}$) run over the set $Z \cap B(\mathbf{0}, p-1)$, such that every $\mathbf{d} \in Z \cap B(\mathbf{0}, p-1)$ occurs exactly $E(\mathbf{d})$ times. Then the $E_{p-1} \times n$ -matrix $\mathcal{D}$, whose rows consists of the words $\mathbf{d}_j$ ($j \in A$) is called the excess matrix of C . The column vectors of $\mathcal{D}$ are the incident vectors of a subset system $B_1, \dots, B_n \subset A$, i.e. for $1 \leq i \leq n$

$$B_i = \{j \in A : \text{the } i\text{th coordinate of } \mathbf{d}_j \text{ is nonzero}\} \subset A.$$

Note that

$$w(\mathbf{d}_j) = p-1 \quad \text{for } j \in A \quad (3.39)$$

by (3.38). Moreover $E_{p-1} > 0$ by (3.35). Thus the excess matrix is nonempty.

We next show, that condition (q) implies a strong structure on the column sets $B_1, \dots, B_n \subset A$ of the excess matrix.

Theorem 31. Assume the propositions of Definition 30 are satisfied. Let $s, r_1, \dots, r_s$ be any integers satisfying $1 \leq s \leq p-1$ and $1 \leq r_1 < \dots < r_s \leq n$. We set $q = p-s$. If C satisfies condition (q), then the column sets B_i , $1 \leq i \leq n$ of the excess matrix satisfy

$$\left| \bigcap_{1 \leq i \leq s} B_{r_i} \right| \equiv 0 \pmod{q}. \quad (3.40)$$

Proof. Let $\mathbf{y} \in \mathbb{F}^n$ denote the unique vector, whose i th coordinate is nonzero if and only if $i \in \{r_1, \dots, r_s\}$ ($1 \leq i \leq n$).

Now if $\mathbf{z} \in B(\mathbf{y}, q-2)$, then $d(\mathbf{z}, \mathbf{0}) \leq d(\mathbf{z}, \mathbf{y}) + d(\mathbf{y}, \mathbf{0}) \leq p-2$ and thus $\mathbf{z} \in B(\mathbf{0}, p-2)$ implying $E(\mathbf{z}) = 0$ by (3.38). Therefore $E_i(\mathbf{y}) = 0$ for $0 \leq i \leq q-2$, which yields

$$E_{q-1}(\mathbf{y}) \equiv 0 \pmod{q} \quad (3.41)$$

by (3.36) (applied to $\mathbf{x} = \mathbf{y}$). If $\mathbf{z} \in Z \cap B(\mathbf{y}, q-1)$, then $d(\mathbf{z}, \mathbf{0}) \leq d(\mathbf{z}, \mathbf{y}) +$

$d(\mathbf{y}, \mathbf{0}) \leq p - 1$ and thus $\mathbf{z} \in Z \cap B(\mathbf{0}, p - 1)$. Therefore

$$\begin{aligned}
E_{q-1}(\mathbf{y}) &= \sum_{\substack{\mathbf{z} \in Z \\ d(\mathbf{z}, \mathbf{y}) = q-1}} E(\mathbf{z}) \\
&= \sum_{\substack{\mathbf{z} \in Z \cap B(\mathbf{0}, p-1) \\ d(\mathbf{z}, \mathbf{y}) = q-1}} E(\mathbf{z}) \\
&= |\{j \in A \mid d(\mathbf{d}_j, \mathbf{y}) = q - 1\}|.
\end{aligned} \tag{3.42}$$

We next show

$$\{j \in A : d(\mathbf{d}_j, \mathbf{y}) = q - 1\} = \bigcap_{1 \leq i \leq s} B_{r_i}. \tag{3.43}$$

Assume first $d(\mathbf{d}_j, \mathbf{y}) = q - 1$ for an integer j with $1 \leq j \leq E_{p-1}$. To show $j \in \bigcap_{1 \leq i \leq s} B_{r_i}$ it suffices to show, that the i th coordinates of $\mathbf{d}_j$ are nonzero whenever $i \in \{r_1, \dots, r_s\}$. Assume to the contrary, that, say the r_1 th coordinate of $\mathbf{d}_j$ is zero. This means that there are at most $s - 1$ nonzero coordinates among the set $\{r_1, \dots, r_s\}$. Since $\mathbf{d}_j$ has exactly $p - 1$ nonzero coordinates, $\mathbf{d}_j$ must have at least $(p - 1) - (s - 1) = q$ nonzero coordinates outside of $\{r_1, \dots, r_s\}$. In these coordinates $\mathbf{y}$ is zero however, implying $d(\mathbf{d}_j, \mathbf{y}) \geq q$, contradicting $d(\mathbf{d}_j, \mathbf{y}) = q - 1$.

Conversely assume $j \in \bigcap_{1 \leq i \leq s} B_{r_i}$. This means that the i th coordinates of $\mathbf{d}_j$ are nonzero whenever $i \in \{r_1, \dots, r_s\}$. Moreover $\mathbf{d}_j$ has exactly $q - 1$ nonzero coordinates outside the set $\{r_1, \dots, r_s\}$. Using the definition of $\mathbf{y}$ we now see, that $\mathbf{y}$ and $\mathbf{d}_j$ exactly differ in these $q - 1$ coordinates, i.e. $d(\mathbf{d}_j, \mathbf{y}) = q - 1$. This completes the proof of (3.43). Now (3.40) follows by (3.41), (3.42) and (3.43). $\square$

Proof of Theorem 29. It suffices to consider the case $\mathbf{x} = \mathbf{0}$. Assume equality holds in (3.35), i.e. $E_{p-1} = |A| = p - 1$. Since any row of the excess matrix $\mathcal{D}$ has weight $p - 1 \geq 2$, $\mathcal{D}$ has at least two nonempty column sets, say $B_1, B_2 \subset A$. Since C satisfies condition (p-1) we may apply Theorem 31 with $s = 1$ and $r_1 = 1$ (resp. $r_1 = 2$) and get $|B_1| \equiv |B_2| \equiv 0 \pmod{p-1}$, implying $B_1 = B_2 = A$ by $|A| = p - 1$. Again by Theorem 31 (with $s = 2$ and $r_1 = 1$, $r_2 = 2$) however we find $|B_1 \cap B_2| \equiv 0 \pmod{p-2}$, since C satisfies condition (p-2). This gives the contradiction $p - 1 = |A| = |B_1| = |B_1 \cap B_2| \equiv 0 \pmod{p-2}$. Therefore $E_{p-1} > p - 1$ and (3.37) follows by (3.34) and Theorem 24. $\square$

In [9] Habsieger explicitly gave congruence properties of type (3.36) for $q \leq 5$ and raised the question, if the conditions $(p-1)$ and $(p-2)$ are always satisfied for C under the propositions of Theorem 29. He wrote:

It is probably true, that a necessary and sufficient condition for the existence of (3.36) (for $q = p-1$ and $q = p-2$) is that p divides $n+1$. Since the applications of ... require explicit congruence properties, we shall not try to prove this characterization.

In the next section we prove the sufficient part of Habsiegers conjecture. Motivated by important structural properties of the excess matrix (see Theorem 31), we show more generally, that condition (q) is satisfied for any binary code of length n and covering radius one, if q and $n+1$ do not have a common odd prime divisor. More precisely we prove the following theorem.

Theorem 32. *Let n, q be integers with $1 \leq q \leq n$ and odd n . Assume q and $n+1$ do not have a common odd prime divisor. Then there are integers $\beta_0, \dots, \beta_{q-2}$ depending only on q and the residue class of $n \pmod{q}$, such that for every binary code $C \subset \mathbb{F}^n$ with covering radius one and each $\mathbf{x} \in \mathbb{F}^n$ we have*

$$\sum_{0 \leq i \leq q-2} \beta_i E_i(\mathbf{x}) + E_{q-1}(\mathbf{x}) \equiv 0 \pmod{q}. \quad (3.44)$$

Note that Habsiegers conjecture follows from Theorem 32 applied to $q = p-2$ resp. $q = p-1$, since by (3.29) and (3.34) p is the least odd prime satisfying $n \equiv -1 \pmod{p}$ and therefore q and $n+1$ do not have a common odd prime divisor.

As the second major question concerning Theorem 29, Habsieger [9] asked whether the estimation $E_{p-1} \geq 2p-1$ could be improved further. Indeed Honkala [17] showed $E_{p-1} \geq 3p-1$. In Habsiegers words:

... the inequality $E_{p-1} \geq 2p-1$ can probably sharpened. One might even get a bound which is quadratic in p .

As a main goal for the rest of this chapter we will answer this question in the affirmative. Under essential use of the structural properties of the excess matrix stated in Theorem 31 together with the congruence properties described in Theorem 32 we will give a proof of the following unconditional result.

Theorem 33. *Assume $C \subset \mathbb{F}^n$ is a binary code of length n and covering radius one with $n \equiv -1 \pmod{p}$ for an odd prime p . Then for every $\mathbf{x} \in \mathbb{F}^n$*

$$E_0(\mathbf{x}) = E_1(\mathbf{x}) = \dots = E_{p-2}(\mathbf{x}) = 0 \quad (3.45)$$

implies

$$E_{p-1}(\mathbf{x}) \geq (p-2)p-1. \quad (3.46)$$

In section 3.7 we will use Theorem 33 to give a "general improvement of the general excess bound" for binary codes with covering radius one.

3.5 Congruence properties for the covering excess function

In this section we prove Theorem 32 and discuss some of its consequences including the outstanding proof of Theorem 26. In this section we always assume, that variables with negative index equal zero.

3.5.1 Some lemmas

For the proof of Theorem 32 we need some lemmas.

Lemma 34. *Let m, n be nonnegative integers with n odd. Then there are integers γ_i ($0 \leq i \leq \frac{n+1}{2}$) satisfying*

$$(n-2i+1)\gamma_i + (2i+2)\gamma_{i+1} \equiv \binom{n+1}{2i+1} \pmod{2^m} \quad (3.47)$$

for $0 \leq i \leq \frac{n-1}{2}$.

Proof. Recursively choose integers u_i ($0 \leq i \leq \frac{n+1}{2}$) satisfying

$$(n-2i)u_{i+1} \equiv 1 - (2i+1)u_i \pmod{2^m}$$

for $0 \leq i \leq \frac{n-1}{2}$ (here u_0 may be chosen arbitrarily). This is possible, since $n-2i$ is odd ($0 \leq i \leq \frac{n-1}{2}$) and therefore $n-2i$ and the modul 2^m do not have a common prime divisor. We now set

$$\gamma_i = u_i \binom{n+1}{2i} \quad \left(0 \leq i \leq \frac{n+1}{2}\right)$$

and find $(0 \leq i \leq \frac{n-1}{2})$

$$\begin{aligned}
& (n-2i+1)\gamma_i + (2i+2)\gamma_{i+1} \\
= & (n-2i+1)u_i \binom{n+1}{2i} + (2i+2)u_{i+1} \binom{n+1}{2i+2} \\
= & (n-2i+1)u_i \binom{n+1}{2i} + (n-2i)u_{i+1} \binom{n+1}{2i+1} \\
\equiv & (n-2i+1)u_i \binom{n+1}{2i} + (1-(2i+1)u_i) \binom{n+1}{2i+1} \pmod{2^m} \\
= & \binom{n+1}{2i+1}.
\end{aligned}$$

□

Lemma 35. *Let m, n be nonnegative integers with n odd. Then there are integers α'_i ($0 \leq i \leq n$) satisfying*

$$(n-i+1)\alpha'_{i-1} + \alpha'_i + (i+1)\alpha'_{i+1} \equiv \binom{n}{i} \pmod{2^m} \quad (3.48)$$

for $0 \leq i \leq n-1$.

Proof. We use Lemma 34 to get integers γ_i ($0 \leq i \leq \frac{n+1}{2}$) satisfying (3.47). The integers α'_i ($0 \leq i \leq n$) are defined recursively. When i is even with $0 \leq i \leq n$ we set

$$\alpha'_i = \gamma_{\frac{i}{2}} - \alpha'_{i-1} \quad (0 \leq i \leq n, i \text{ even}). \quad (3.49)$$

When i is odd with $0 \leq i \leq n$ we choose any integer α'_i satisfying

$$i\alpha'_i \equiv \binom{n}{i-1} - (n-i+2)\alpha'_{i-2} - \alpha'_{i-1} \pmod{2^m} \quad (0 \leq i \leq n, i \text{ odd}). \quad (3.50)$$

This is possible, since i and 2^m do not have a common prime divisor if i is odd. When i is even with $0 \leq i \leq n-1$, then (3.48) follows from (3.50)

(with $i + 1$ instead of i). When i is odd with $0 \leq i \leq n - 1$ we have

$$\begin{aligned}
& (n - i + 1)\alpha'_{i-1} + \alpha'_i + (i + 1)\alpha'_{i+1} \\
&= (n - i + 2)(\alpha'_{i-2} + \alpha'_{i-1}) + (i + 1)(\alpha'_i + \alpha'_{i+1}) \\
&\quad - ((n - i + 2)\alpha'_{i-2} + \alpha'_{i-1} + i\alpha'_i) \\
&= \left(n - 2\frac{i-1}{2} + 1\right) \gamma_{\frac{i-1}{2}} + \left(2\frac{i-1}{2} + 2\right) \gamma_{\frac{i+1}{2}} \quad \text{by (3.49)} \\
&\quad - ((n - i + 2)\alpha'_{i-2} + \alpha'_{i-1} + i\alpha'_i) \\
&\equiv \binom{n+1}{i} - \binom{n}{i-1} \pmod{2^m} \quad \text{by (3.47), (3.50)} \\
&= \binom{n}{i}
\end{aligned}$$

and (3.48) holds again. $\square$

Lemma 36. *Assume n, u are nonnegative odd integers, such that $n+1$ and u do not have a common prime divisor. Then there are integers α''_i ($0 \leq i \leq n$) satisfying*

$$(n - i + 1)\alpha''_{i-1} + \alpha''_i + (i + 1)\alpha''_{i+1} \equiv \binom{n}{i} \pmod{u} \quad (3.51)$$

for $0 \leq i \leq n - 1$.

Proof. Choose an integer α satisfying

$$\alpha(n + 1) \equiv 1 \pmod{u}. \quad (3.52)$$

This is possible, since u and $n + 1$ are assumed to be without common prime divisor. We set

$$\alpha''_i = \alpha \binom{n}{i} \quad (0 \leq i \leq n)$$

and have $(0 \leq i \leq n-1)$

$$\begin{aligned}
& (n-i+1)\alpha''_{i-1} + \alpha''_i + (i+1)\alpha''_{i+1} \\
&= \alpha \left((n-i+1) \binom{n}{i-1} + \binom{n}{i} + (i+1) \binom{n}{i+1} \right) \\
&= \alpha \left(i \binom{n}{i} + \binom{n}{i} + (n-i) \binom{n}{i} \right) \\
&= \alpha(n+1) \binom{n}{i} \\
&\equiv \binom{n}{i} \pmod{u} \quad \text{by (3.52).}
\end{aligned}$$

□

Lemma 37. *Let n, q be positive integers with n odd. Moreover assume, that $n+1$ and q do not have a common odd prime divisor. Then there are integers α_i ($0 \leq i \leq n$) satisfying*

$$(n-i+1)\alpha_{i-1} + \alpha_i + (i+1)\alpha_{i+1} \equiv \binom{n}{i} \pmod{q} \quad (3.53)$$

for $0 \leq i \leq n-1$.

Proof. Write

$$q = 2^m u \quad (3.54)$$

with $m \geq 0$ and odd u . Then the propositions of Lemma 35 and Lemma 36 are satisfied and we get integers α'_i, α''_i ($0 \leq i \leq n$) satisfying (3.48) and (3.51). Now by the well-known chinese remainder theorem there are integers α_i ($0 \leq i \leq n$) satisfying

$$\begin{aligned}
\alpha_i &\equiv \alpha'_i \pmod{2^m}, \\
\alpha_i &\equiv \alpha''_i \pmod{u}
\end{aligned}$$

for $0 \leq i \leq n$, since 2^m and u do not have a common prime divisor. By (3.48) and (3.51) this implies

$$\begin{aligned}
(n-i+1)\alpha_{i-1} + \alpha_i + (i+1)\alpha_{i+1} &\equiv \binom{n}{i} \pmod{2^m}, \\
(n-i+1)\alpha_{i-1} + \alpha_i + (i+1)\alpha_{i+1} &\equiv \binom{n}{i} \pmod{u}
\end{aligned}$$

for $0 \leq i \leq n-1$ and (3.53) follows by (3.54). □

Lemma 38. *Let n, q be positive integers with n odd and $q \geq 2$. Then there are integers β_i ($0 \leq i \leq q-1$) depending only on q and the residue class of $n \pmod{q}$ satisfying*

$$(n-i)\beta_{i+1} + \beta_i + i\beta_{i-1} \equiv 0 \pmod{q} \quad (3.55)$$

for $0 \leq i \leq q-2$ and

$$\beta_{q-1} = \beta_{q-2} = 1. \quad (3.56)$$

Proof. Note that, since n is odd, there exists a suitable odd positive integer $r \leq 2q-1$ with $n \equiv r \pmod{q}$. For $0 \leq i \leq q-1$ we set

$$\begin{aligned} \beta &= \frac{r+3}{2}, \\ k_i &= \left\lfloor \frac{q-i-1}{2} \right\rfloor, \\ \beta_i &= \binom{\beta + k_i - 1}{k_i}. \end{aligned}$$

Of course β, β_i ($0 \leq i \leq q-1$) are integers, because r is odd and $k_i \geq 0$ for $0 \leq i \leq q-1$. Now (3.56) is satisfied by $k_{q-1} = k_{q-2} = 0$. It remains to show (3.55). We consider two cases.

Case I. $0 \leq i \leq q-2$ with $q-i$ even.

Then $k_i = k_{i+1} = k_{i-1} - 1$ (for $i > 0$) and thus

$$\begin{aligned} & (n-i)\beta_{i+1} + \beta_i + i\beta_{i-1} \\ \equiv & (r-i) \binom{\beta + k_{i+1} - 1}{k_{i+1}} + \binom{\beta + k_i - 1}{k_i} + i \binom{\beta + k_{i-1} - 1}{k_{i-1}} \pmod{q} \\ = & (r-i+1) \binom{\beta + k_{i+1} - 1}{k_{i+1}} + i \binom{\beta + k_{i-1} - 1}{k_{i-1}} \\ \equiv & (r+q-i+1) \binom{\beta + k_{i+1} - 1}{k_{i+1}} - (q-i) \binom{\beta + k_{i-1} - 1}{k_{i-1}} \pmod{q} \\ = & 2(\beta + k_{i+1}) \binom{\beta + k_{i+1} - 1}{k_{i+1}} - 2(k_{i+1} + 1) \binom{\beta + k_{i+1}}{k_{i+1} + 1} \\ = & 0 \end{aligned}$$

and (3.55) follows in case I. Note, that this holds in the case $i = 0$ too.

Case II. $0 \leq i \leq q - 2$ with $q - i$ odd.

Then $k_{i-1} = k_i = k_{i+1} + 1$ (for $i > 0$) and thus

$$\begin{aligned}
& (n - i)\beta_{i+1} + \beta_i + i\beta_{i-1} \\
\equiv & (r - i) \binom{\beta + k_{i+1} - 1}{k_{i+1}} + (i + 1) \binom{\beta + k_i - 1}{k_i} \pmod{q} \\
\equiv & (r + q - i) \binom{\beta + k_{i+1} - 1}{k_{i+1}} - (q - i - 1) \binom{\beta + k_i - 1}{k_i} \pmod{q} \\
= & 2(\beta + k_i - 1) \binom{\beta + k_i - 2}{k_i - 1} - 2k_i \binom{\beta + k_i - 1}{k_i} \\
= & 0
\end{aligned}$$

and (3.55) follows again. Note, that the integers β_i ($0 \leq i \leq q - 1$) only depend on q and r , the residue class of $n \pmod{q}$. $\square$

3.5.2 Proof of Theorem 32

Using (3.28) one immediately sees, that Theorem 32 follows from Theorem 39, a purely number theoretic version.

Theorem 39. *Let n, q be integers with $1 \leq q \leq n$ and odd n , such that q and $n + 1$ do not have a common odd prime divisor. Then there are integers $\beta_0, \dots, \beta_{q-2}$ depending only on q and the residue class of $n \pmod{q}$ satisfying*

$$\sum_{0 \leq i \leq q-2} \beta_i E_i + E_{q-1} \equiv 0 \pmod{q} \quad (3.57)$$

with

$$E_i = E_i(A_0, \dots, A_q) = (n + 1 - i)A_{i-1} + A_i + (i + 1)A_{i+1} - \binom{n}{i} \quad (3.58)$$

for $0 \leq i \leq q - 1$ and any integers $A_0, \dots, A_q$.

Proof. There is nothing to prove when $q = 1$, so we may assume $q \geq 2$. The propositions of Lemma 38 are satisfied. Therefore there exist integers β_i ($0 \leq i \leq q - 1$) depending only on q and the residue class of $n \pmod{q}$ satisfying (3.55) and (3.56) for $0 \leq i \leq q - 2$.

Inserting (3.58) we find

$$\begin{aligned}
& \sum_{0 \leq i \leq q-2} \beta_i E_i + E_{q-1} \tag{3.59} \\
&= \sum_{0 \leq i \leq q-1} \beta_i E_i \quad \text{by (3.56)} \\
&= \sum_{0 \leq i \leq q-1} \beta_i \left((n+1-i)A_{i-1} + A_i + (i+1)A_{i+1} - \binom{n}{i} \right) \\
&= \sum_{0 \leq i \leq q-2} ((n-i)\beta_{i+1} + \beta_i + i\beta_{i-1})A_i - \sum_{0 \leq i \leq q-1} \beta_i \binom{n}{i} \\
&\quad + \beta_{q-1}A_{q-1} + \beta_{q-2}(q-1)A_{q-1} + \beta_{q-1}qA_q \\
&\equiv - \sum_{0 \leq i \leq q-1} \beta_i \binom{n}{i} \pmod{q}
\end{aligned}$$

by (3.55) and (3.56). We now use Lemma 37. The propositions are satisfied. Therefore there exist integers α_i ($0 \leq i \leq n$) satisfying (3.53) for $0 \leq i \leq n-1$. We thus have

$$\begin{aligned}
\sum_{0 \leq i \leq q-1} \beta_i \binom{n}{i} &\equiv \sum_{0 \leq i \leq q-1} \beta_i ((n-i+1)\alpha_{i-1} + \alpha_i + (i+1)\alpha_{i+1}) \pmod{q} \\
&= \sum_{0 \leq i \leq q-2} \alpha_i ((n-i)\beta_{i+1} + \beta_i + i\beta_{i-1}) \\
&\quad + \alpha_{q-1}\beta_{q-1} + (q-1)\alpha_{q-1}\beta_{q-2} + q\alpha_q\beta_{q-1} \\
&\equiv 0 \pmod{q}
\end{aligned}$$

by (3.55) and (3.56). Inserting this in (3.59) we get (3.57). $\square$

3.5.3 Proof of Theorem 26

We start to show by induction, that there are integers $A_0, \dots, A_{p-1}$ satisfying (3.33) for $0 \leq i \leq p-2$. We set $A_0 = 0$, $A_1 = 1$ and (3.33) is satisfied for $i = 0$. Now let q be an integer with $2 \leq q \leq p-1$ and assume $A_0, \dots, A_{q-1}$ are integers satisfying (3.33) for $0 \leq i \leq q-2$. Choose an integer A_q such that $|E_{q-1}|$ is minimal. We apply Theorem 39. Apparently q and $n+1$ do not have a common odd prime divisor, since $q < p$ and p is assumed to be

the least odd prime divisor of $n + 1$. Therefore the propositions of Theorem 39 are satisfied. Thus we may apply (3.57) and get $E_{q-1} \equiv 0 \pmod{q}$ by $E_i = 0$ for $0 \leq i \leq q - 2$. Since the coefficient of A_q in (3.33) for $i = q - 1$ is q we have $E_{q-1} = 0$ because A_q is assumed to be chosen in such a way, that $|E_{q-1}|$ is minimal. Thus (3.33) holds for $i = q - 1$ too and by induction on q there exist integers $A_0, \dots, A_{p-1}$ satisfying (3.33) for $0 \leq i \leq p - 2$. It remains to show, that $A_0, \dots, A_{p-1}$ are nonnegative. By induction we show

$$0 \leq A_i \leq \frac{1}{i} \binom{n}{i-1} \quad (3.60)$$

for $1 \leq i \leq p - 1$. By $A_0 = 0$, $A_1 = 1$ and (3.33) we have $A_2 = (n - 1)/2$. Therefore (3.60) is satisfied for $i = 1, 2$. Now assume $3 \leq q \leq p - 1$ and (3.60) holds for $1 \leq i \leq q - 1$. We show

$$0 \leq A_q \leq \frac{1}{q} \binom{n}{q-1}. \quad (3.61)$$

The inequality at the right-hand side of (3.61) immediately follows from (3.33) (for $i = q - 1$) and $A_{q-2} \geq 0$, $A_{q-1} \geq 0$ (by the left-hand side of (3.60) for $i = q - 2, q - 1$). To show the left-hand side of (3.61) we have

$$\begin{aligned} qA_q &= \binom{n}{q-1} - (n - q + 2)A_{q-2} - A_{q-1} && \text{by 3.33} \\ &\geq \binom{n}{q-1} - \frac{n - q + 2}{q - 2} \binom{n}{q-3} - \frac{1}{q-1} \binom{n}{q-2} && \text{by 3.60} \\ &= \left(1 - \frac{q-1}{n-q+3} - \frac{1}{n-q+2}\right) \binom{n}{q-1} \\ &\geq \left(1 - \frac{q}{n-q+2}\right) \binom{n}{q-1} \\ &\geq \left(1 - \frac{p}{n-p+2}\right) \binom{n}{q-1} && \text{by } q \leq p \\ &\geq 0 \end{aligned}$$

by $p \leq \frac{n+1}{2}$ (since p is an odd prime dividing the even number $n + 1$). Therefore the inequality at the left-hand side of (3.61) follows. By induction on q now (3.60) follows for $1 \leq i \leq p - 1$. This completes the proof of Theorem 26.

3.5.4 The main property of the excess matrix

Again let $C \subset \mathbb{F}^n$ be a binary code of odd length n and covering radius one with $n \equiv -1 \pmod{p}$ for an odd prime p . Assume

$$E_0 = E_1 = \dots = E_{p-2} = 0 \quad (3.62)$$

with $E_i = E_i(\mathbf{0})$ for $i \geq 0$. Remember that the excess matrix $\mathcal{D}$ of C and their column sets $B_1, \dots, B_n \subset A$ were defined in Definition 30. We now use Theorem 31 and Theorem 32 to prove the following main theorem on the excess matrix.

Theorem 40. *The column sets $B_1, \dots, B_n \subset A$ of the excess matrix $\mathcal{D}$ always satisfy*

$$|\{1 \leq i \leq n \mid a \in B_i\}| = p - 1 \quad \text{whenever } a \in A \quad (3.63)$$

and

$$\left| \bigcap_{1 \leq i \leq s} B_{r_i} \right| \equiv 0 \pmod{p-s} \quad (3.64)$$

whenever $1 \leq s \leq p-1$ and $r_1, \dots, r_s$ are integers with $1 \leq r_1 < r_2 < \dots < r_s \leq n$.

Proof. (3.63) follows directly from (3.39), since each row of the excess matrix is a word $\mathbf{d}_j$ for a suitable $j \in A$. Concerning (3.64), let $1 \leq s \leq p-1$ and $r_1, \dots, r_s$ be integers with $1 \leq r_1 < r_2 < \dots < r_s \leq n$. We set $q = p - s$. Now q and $n+1$ cannot have a common odd prime divisor $p' \leq q < p$, since then $n \equiv -1 \pmod{p'}$ and thus $E_0 + E_1 + \dots + E_{p'-1} \equiv -1 \pmod{p'}$ by Theorem 24, contradicting (3.62). Therefore the propositions of Theorem 32 are satisfied. An application of Theorem 32 now yields, that C satisfies condition (q) (see definition 28) and (3.64) follows by Theorem 31. $\square$

As the next major step towards the proof of Theorem 33 we proceed with a detailed study of subset systems satisfying (3.63) and (3.64) in the next section.

3.6 Congruence set systems and their resolvability

Definition 41. *Let A be a finite nonempty set and p an integer with $p \geq 2$. A p -congruence set system on A is a sequence of (not necessarily distinct)*

subsets $B_1, \dots, B_n \subset A$ satisfying

$$|\{1 \leq i \leq n \mid a \in B_i\}| = p - 1 \quad (3.65)$$

for every $a \in A$ and

$$\left| \bigcap_{1 \leq i \leq s} B_{r_i} \right| \equiv 0 \pmod{p-s} \quad (3.66)$$

whenever $1 \leq s \leq p-1$ and $r_1, \dots, r_s$ are integers with $1 \leq r_1 < r_2 < \dots < r_s \leq n$.

Note, that p is not required to be prime in this section. Moreover (3.65) implies $1 \leq p-1 \leq n$, because A is supposed to be nonempty.

Let us use $p-CSS(A; B_1, \dots, B_n)$ as an abbreviation for the statement, that $p \geq 2$, A is a nonempty finite set and the subsets $B_1, \dots, B_n \subset A$ form a p -congruence set system on A .

3.6.1 Simple properties of congruence set systems

Lemma 42. *If $p \geq 3$, then $p-CSS(A; B_1, \dots, B_n)$ implies*

$$(p-1)-CSS(B_j; B_1 \cap B_j, \dots, B_{j-1} \cap B_j, B_{j+1} \cap B_j, \dots, B_n \cap B_j)$$

whenever $1 \leq j \leq n$ with $B_j \neq \emptyset$.

Proof. Using $p-CSS(A; B_1, \dots, B_n)$ we find

$$|\{1 \leq i \leq n \mid i \neq j, a \in B_i \cap B_j\}| = |\{1 \leq i \leq n \mid a \in B_i\}| - |\{j\}| = p - 2$$

whenever $a \in B_j$ and

$$\left| \bigcap_{1 \leq i \leq s} (B_{r_i} \cap B_j) \right| = \left| B_j \cap \bigcap_{1 \leq i \leq s} B_{r_i} \right| \equiv 0 \pmod{p-1-s}$$

whenever $1 \leq s \leq p-2$ and $r_1, \dots, r_s$ are integers with $1 \leq r_1 < \dots < r_s \leq n$, $r_i \neq j$ ($1 \leq i \leq s$). From this Lemma 42 immediately follows. $\square$

Lemma 43. *Assume $A'' \subset A$ with $|A''| < |A|$. Then $p-CSS(A; B_1, \dots, B_n)$ and $p-CSS(A''; B_1 \cap A'', \dots, B_n \cap A'')$ imply*

$$p-CSS(A \setminus A''; B_1 \setminus A'', \dots, B_n \setminus A'').$$

Proof. By $p - CSS(A; B_1, \dots, B_n)$ we find

$$|\{1 \leq i \leq n \mid a \in B_i \setminus A''\}| = |\{1 \leq i \leq n : a \in B_i\}| = p - 1$$

for each $a \in A \setminus A''$. Moreover, if $1 \leq s \leq p - 1$ and $r_1, \dots, r_s$ are integers with $1 \leq r_1 < \dots < r_s \leq n$, then

$$\begin{aligned} \left| \bigcap_{1 \leq i \leq s} (B_{r_i} \setminus A'') \right| &= \left| \bigcap_{1 \leq i \leq s} B_{r_i} \setminus \bigcap_{1 \leq i \leq s} (B_{r_i} \cap A'') \right| \\ &= \left| \bigcap_{1 \leq i \leq s} B_{r_i} \right| - \left| \bigcap_{1 \leq i \leq s} (B_{r_i} \cap A'') \right| \\ &\equiv 0 \pmod{p - s} \end{aligned}$$

by $p - CSS(A; B_1, \dots, B_n)$ and $p - CSS(A''; B_1 \cap A'', \dots, B_n \cap A'')$ and Lemma 43 follows. $\square$

Lemma 44. $p - CSS(A; B_1, \dots, B_n)$ implies

$$(p - 1)|A| = \sum_{1 \leq i \leq n} |B_i|. \quad (3.67)$$

Proof. By (3.65) we have

$$(p - 1)|A| = (p - 1) \sum_{a \in A} 1 = \sum_{a \in A} \sum_{\substack{1 \leq i \leq n \\ a \in B_i}} 1 = \sum_{1 \leq i \leq n} |B_i|.$$

$\square$

In the next lemma we give a simple, but important example of a congruence set system.

Lemma 45. Assume $p \geq 2$ and $A = \{a_1, \dots, a_p\}$ with $|A| = p$. Then

$$p - CSS(A; A \setminus \{a_1\}, \dots, A \setminus \{a_p\}).$$

Proof. For every $a \in A$ there exists a unique number j with $1 \leq j \leq p$ and $a = a_j$. Therefore

$$|\{1 \leq i \leq p \mid a \in A \setminus \{a_i\}\}| = |\{1 \leq i \leq p \mid i \neq j\}| = p - 1.$$

Moreover whenever $1 \leq s \leq p - 1$ and $r_1, \dots, r_s$ are integers with $1 \leq r_1 < \dots < r_s \leq p$ then

$$\left| \bigcap_{1 \leq i \leq s} (A \setminus \{a_{r_i}\}) \right| = |A \setminus \{a_{r_1}, \dots, a_{r_s}\}| = p - s \equiv 0 \pmod{p - s}$$

and Lemma 45 follows. $\square$

Lemma 45 may be used to construct more examples of congruence set systems. Let $\mathcal{P}_k(A)$ denote the set of the k -element subsets of A . Assume A is a finite nonempty set and p an integer with $p \geq 2$. We call a sequence of (not necessarily distinct) subsets $B_1, \dots, B_n \subset A$ p -resolvable on A , if there is a partition $A_1 \cup \dots \cup A_t = A$ of A in pairwise disjoint subsets $A_j \subset A$ ($1 \leq j \leq t$) satisfying

$$|A_j| = p,$$

$$|\{1 \leq i \leq n \mid B_i \cap A_j \neq \emptyset\}| = p$$

and

$$\{B_i \cap A_j \mid 1 \leq i \leq n \text{ with } B_i \cap A_j \neq \emptyset\} = \mathcal{P}_{p-1}(A_j)$$

whenever $1 \leq j \leq t$. Using Lemma 45 it is not difficult to see, that $p - CSS(A; B_1, \dots, B_n)$ holds whenever the sequence $B_1, \dots, B_n \subset A$ is p -resolvable on A . The main object of the next section is to show a partial converse.

3.6.2 The classification of congruence set systems on small sets

In the next theorem we give a complete characterization for congruence set systems on a set A with $|A| \leq (p - 3)p$.

Theorem 46. *Assume $p - CSS(A; B_1, \dots, B_n)$ with*

$$|A| \leq (p - 3)p. \tag{3.68}$$

Then the sequence $B_1, \dots, B_n \subset A$ is p -resolvable on A . Especially $|A|$ is divisible by p .

Proof. Without loss of generality we may assume $B_i \neq \emptyset$ for $1 \leq i \leq n$, since empty sets occuring in the sequence $B_1, \dots, B_n$ do neither influence the property of being a p -congruence set system nor the notion of p -resolvability. The propositions of Theorem 46 are not satisfied for $p \leq 3$, so we may assume $p \geq 4$ (and thus $n \geq 3$). Theorem 46 follows by induction on p and $|A|$, if we are able to show, that the sequence $B_1, \dots, B_n \subset A$ is p -resolvable on A under the assumption of the following statements (i) and (ii).

(i) If $2 \leq p' < p$, $p' - CSS(A'; B'_1, \dots, B'_{n'})$ and

$$|A'| \leq (p' - 3)p', \quad (3.69)$$

then the sequence $B'_1, \dots, B'_{n'}$ is p' -resolvable on A' .

(ii) If $|A'| < |A|$ and $p - CSS(A'; B'_1, \dots, B'_{n'})$, then the sequence $B'_1, \dots, B'_{n'}$ is p -resolvable on A' .

We first show, that it suffices to prove the existence of a subset $A'' \subset A$ satisfying

$$|A''| = p \quad (3.70)$$

and

$$\{B_i \cap A'' \mid 1 \leq i \leq n \text{ with } B_i \cap A'' \neq \emptyset\} = \mathcal{P}_{p-1}(A''). \quad (3.71)$$

Note that from (3.65), (3.71) and $|\mathcal{P}_{p-1}(A'')| = p$ follows

$$|\{1 \leq i \leq n \mid B_i \cap A'' \neq \emptyset\}| = p. \quad (3.72)$$

Therefore, if $A'' = A$, then the sequence $B_1, \dots, B_n$ is p -resolvable on A with the only resolution class A'' . If $|A''| < |A|$, then (3.71), (3.72) and an application of Lemma 45 with $A = A''$ immediately imply $p - CSS(A''; B_1 \cap A'', \dots, B_n \cap A'')$. Now by Lemma 43 we have $p - CSS(A \setminus A''; B_1 \setminus A'', \dots, B_n \setminus A'')$. We thus may apply statement (ii) with $n' = n$, $A' = A \setminus A''$ and $B'_i = B_i \setminus A''$ ($1 \leq i \leq n$). We find that the sequence $B_1 \setminus A'', \dots, B_n \setminus A''$ is p -resolvable on $A \setminus A''$. This means that there is a partition $A_1 \cup \dots \cup A_t = A \setminus A''$ of $A \setminus A''$ in pairwise disjoint subsets $A_j \subset A \setminus A''$ ($1 \leq j \leq t$) satisfying

$$|A_j| = p,$$

$$|\{1 \leq i \leq n \mid (B_i \setminus A'') \cap A_j \neq \emptyset\}| = p$$

and

$$\{(B_i \setminus A'') \cap A_j \mid 1 \leq i \leq n \text{ with } (B_i \setminus A'') \cap A_j \neq \emptyset\} = \mathcal{P}_{p-1}(A_j)$$

for $1 \leq j \leq t$. If we now set $A_{t+1} = A''$, then this together with (3.70), (3.71) and (3.72) implies, that $A_1 \cup \dots \cup A_t \cup A_{t+1} = A$ with pairwise disjoint subsets $A_j \subset A$ ($1 \leq j \leq t+1$) satisfying

$$|A_j| = p,$$

$$|\{1 \leq i \leq n \mid B_i \cap A_j \neq \emptyset\}| = p$$

and

$$\{B_i \cap A_j \mid 1 \leq i \leq n \text{ with } B_i \cap A_j \neq \emptyset\} = \mathcal{P}_{p-1}(A_j)$$

for $1 \leq j \leq t+1$. This means, that the sequence $B_1, \dots, B_n$ is p -resolvable on A . Therefore the proof of Theorem 46 is complete, if we can show that there exists a subset $A'' \subset A$ satisfying (3.70) and (3.71). Let us consider two cases.

Case I. $|B_i| \geq (p-3)(p-1)$ for $1 \leq i \leq n$.

By (3.66) we have $|B_i| \equiv 0 \pmod{p-1}$ for $1 \leq i \leq n$. This implies, that indeed in case I we have

$$|B_i| = (p-3)(p-1) \quad \text{for } 1 \leq i \leq n, \quad (3.73)$$

since otherwise by (3.68) we would have the contradiction $|B_i| \geq (p-2)(p-1) > (p-3)p \geq |A| \geq |B_i|$ ($1 \leq i \leq n$). Again by (3.66) we have $|B_1 \cap B_2| \equiv 0 \pmod{p-2}$ (remember $p \geq 4$ and $n \geq 3$). This implies $|B_1 \cap B_2| \leq (p-3)(p-2)$, since otherwise by (3.73) we would have the contradiction $|B_1 \cap B_2| \geq (p-2)^2 > (p-3)(p-1) = |B_1| \geq |B_1 \cap B_2|$. We therefore have

$$\begin{aligned} |A| &\geq |B_1 \cup B_2| = |B_1| + |B_2| - |B_1 \cap B_2| \\ &\geq 2(p-3)(p-1) - (p-3)(p-2) \\ &= (p-3)p, \end{aligned}$$

which implies

$$|A| = (p-3)p \quad (3.74)$$

by (3.68). Now (3.73) and (3.74) inserted in (3.67) immediately yields

$$n = p. \quad (3.75)$$

From (3.73), (3.74) and $p \geq 4$ follows $A \setminus B_i \neq \emptyset$ for $1 \leq i \leq p$. Therefore we may choose $a_i \in A \setminus B_i$ ($1 \leq i \leq p$). We set

$$A'' = \{a_1, \dots, a_p\}.$$

We next show

$$|A''| = p. \quad (3.76)$$

Otherwise at least two of the elements $a_1, \dots, a_p$ must be equal, say $a_1 = a_2$. Then by definition $a_1 \notin B_1 \cup B_2$. Therefore by (3.75)

$$|\{1 \leq i \leq n \mid a_1 \in B_i\}| \leq |\{3 \leq i \leq p \mid a_1 \in B_i\}| \leq p - 2$$

contradicting (3.65).

We now show

$$B_i \cap A'' = A'' \setminus \{a_i\} \quad (1 \leq i \leq p). \quad (3.77)$$

Fix i_0 with $1 \leq i_0 \leq p$. We have $B_{i_0} \cap A'' \subset A'' \setminus \{a_{i_0}\}$ by $a_{i_0} \notin B_{i_0}$. It remains to show $A'' \setminus \{a_{i_0}\} \subset B_{i_0} \cap A''$. Assume $a \in A'' \setminus \{a_{i_0}\}$, say $a = a_{i_1}$ with $1 \leq i_1 \leq p$, $i_1 \neq i_0$. Then $a_{i_1} \in B_{i_0}$, since otherwise by (3.75) and $a_{i_1} \notin B_{i_1}$

$$|\{1 \leq i \leq n \mid a_{i_1} \in B_i\}| \leq |\{1 \leq i \leq p \mid i \neq i_0, i \neq i_1\}| \leq p - 2$$

contradicting (3.65). Therefore $A'' \setminus \{a_{i_0}\} \subset B_{i_0} \cap A''$ and (3.77) follows. By (3.75) and (3.77) we now have

$$\begin{aligned} \{B_i \cap A'' \mid 1 \leq i \leq n \text{ with } B_i \cap A'' \neq \emptyset\} &= \{A'' \setminus \{a_1\}, \dots, A'' \setminus \{a_p\}\} \\ &= \mathcal{P}_{p-1}(A''). \end{aligned}$$

Thus by (3.76) the subset $A'' \subset A$ satisfies (3.70) and (3.71). This completes the proof of Theorem 46 in case I.

Case II. There exists an integer i_0 with $1 \leq i_0 \leq n$ and

$$|B_{i_0}| < (p - 3)(p - 1).$$

Without loss of generality we may assume $i_0 = 1$. Using $|B_1| \equiv 0 \pmod{p - 1}$ and $B_1 \neq \emptyset$ we thus may assume

$$|B_1| = c(p - 1) \leq (p - 4)(p - 1) \quad (3.78)$$

with a suitable integer c with $1 \leq c \leq p - 4$. Especially in case II we have $p \geq 5$ and $n \geq 4$. By Lemma 42 we now have

$$(p - 1) - CSS(B_1; B_2 \cap B_1, \dots, B_n \cap B_1).$$

From (3.78) follows, that (3.69) is satisfied for $A' = B_1$, $n' = n - 1$, $B'_i = B_{i+1} \cap B_1$ ($1 \leq i \leq n'$) and $p' = p - 1$. Therefore by (i) we know, that the sequence $B_2 \cap B_1, \dots, B_n \cap B_1$ is $(p - 1)$ -resolvable on B_1 . Thus (use (3.78)) there exists a partition $B_1 = B_1^* \cup \dots \cup B_c^*$ in pairwise disjoint sets $B_j^* \subset B_1$ ($1 \leq j \leq c$) satisfying

$$|B_j^*| = p - 1, \quad (3.79)$$

$$|\{2 \leq i \leq n \mid B_i \cap B_j^* \neq \emptyset\}| = p - 1$$

and

$$\{B_i \cap B_j^* \mid 2 \leq i \leq n \text{ with } B_i \cap B_j^* \neq \emptyset\} = \mathcal{P}_{p-2}(B_j^*) \quad (3.80)$$

for $1 \leq j \leq c$. For the case $j = 1$ without loss of generality we further may assume

$$\begin{aligned} \{B_i \cap B_1^* \mid 2 \leq i \leq n \text{ with } B_i \cap B_1^* \neq \emptyset\} &= \{B_2 \cap B_1^*, \dots, B_p \cap B_1^*\} \\ &= \mathcal{P}_{p-2}(B_1^*), \\ B_i \cap B_1^* &= \emptyset \quad \text{for } p < i \leq n. \end{aligned} \quad (3.81)$$

We next show, that the proof of Theorem 46 is complete, if we may prove the existence of $a \in A \setminus B_1$ satisfying

$$a \in B_i \quad (2 \leq i \leq p). \quad (3.82)$$

Note that by (3.65) and (3.82) we have

$$a \notin B_i \quad (p < i \leq n). \quad (3.83)$$

If we set

$$A'' = B_1^* \cup \{a\}$$

then by (3.79) and $a \notin B_1^* \subset B_1$ we have

$$|A''| = p. \quad (3.84)$$

Using $B_1 \cap A'' = B_1^*$ now from (3.81), (3.82) and (3.83) follows

$$\begin{aligned}
& \{B_i \cap A'' \mid 1 \leq i \leq n \text{ with } B_i \cap A'' \neq \emptyset\} \\
&= \{B_1 \cap A'', \dots, B_p \cap A''\} \\
&= \{B_1^* \cup \{(B_2 \cap B_1^*) \cup \{a\}, \dots, (B_p \cap B_1^*) \cup \{a\}\}\} \\
&= \{B_1^* \cup \{D \cup \{a\} \mid D \subset B_1^*, |D| = p-2\}\} \\
&= \mathcal{P}_{p-1}(A'').
\end{aligned}$$

Therefore by (3.84) the subset $A'' \subset A$ satisfies (3.70) and (3.71) and Theorem 46 follows, if we show, that there exists $a \in A \setminus B_1$ satisfying (3.82). We consider two subcases.

$$\text{Case A. } |B_i| \geq (p-3)(p-1) \quad (2 \leq i \leq p). \quad (3.85)$$

In this case by (3.79) and (3.80) we find $B_j^* \setminus B_i \neq \emptyset$ ($1 \leq j \leq c$, $2 \leq i \leq n$) and thus

$$|B_1 \setminus B_i| = \sum_{1 \leq j \leq c} |B_j^* \setminus B_i| \geq c \quad (2 \leq i \leq n). \quad (3.86)$$

Moreover by (3.68), (3.85) and $p \geq 5$ we have

$$(|A| - |B_i|)(p-2) \leq (p-3)(p-2) < (p-3)(p-1) \leq |B_i|$$

for $2 \leq i \leq p$. Addition of $|A| - |B_i|$ yields

$$(|A| - |B_i|)(p-1) < |A| \quad (2 \leq i \leq p). \quad (3.87)$$

Using $(A \setminus B_1) \setminus B_i = (A \setminus B_i) \setminus (B_1 \setminus B_i)$ for $2 \leq i \leq p$ we now find

$$\begin{aligned}
\left| \bigcup_{2 \leq i \leq p} ((A \setminus B_1) \setminus B_i) \right| &\leq \sum_{2 \leq i \leq p} (|A \setminus B_i| - |B_1 \setminus B_i|) \\
&< \frac{|A|}{p-1}(p-1) - c(p-1) \quad \text{by (3.86) and (3.87)} \\
&= |A \setminus B_1| \quad \text{by (3.78)}.
\end{aligned}$$

Therefore there exists $a \in A \setminus B_1$ with $a \notin (A \setminus B_1) \setminus B_i$ for every i with $2 \leq i \leq p$ and (3.82) follows in subcase A.

Case B. There exists j_0 with $2 \leq j_0 \leq p$ and

$$|B_{j_0}| < (p-3)(p-1).$$

Like at the beginning of case II without loss of generality we may assume $j_0 = 2$ and

$$|B_2| = d(p-1)$$

with a suitable integer d with $1 \leq d \leq p-4$. Likewise we may assume that the sequence $B_1 \cap B_2, B_3 \cap B_2, \dots, B_n \cap B_2$ is $(p-1)$ -resolvable on B_2 . Therefore there is a partition $B_2 = B_1^{**} \cup \dots \cup B_d^{**}$ in pairwise disjoint subsets $B_j^{**} \subset B_2$ ($1 \leq j \leq d$) satisfying

$$|B_j^{**}| = p-1,$$

$$|\{1 \leq i \leq n, i \neq 2 \mid B_i \cap B_j^{**} \neq \emptyset\}| = p-1 \quad (3.88)$$

and

$$\{B_i \cap B_j^{**} \mid 1 \leq i \leq n, i \neq 2 \text{ with } B_i \cap B_j^{**} \neq \emptyset\} = \mathcal{P}_{p-2}(B_j^{**}) \quad (3.89)$$

for $1 \leq j \leq d$. By (3.81) we have

$$|B_2 \cap B_1^*| = p-2. \quad (3.90)$$

From this follows, that there exists an integer j_1 with $1 \leq j_1 \leq d$ satisfying

$$|B_{j_1}^{**} \cap B_1^*| \geq 2. \quad (3.91)$$

Otherwise we would have $|B_j^{**} \cap B_1^*| \leq 1$ for $1 \leq j \leq d$ and thus

$$|B_2 \cap B_1^*| = \sum_{1 \leq j \leq d} |B_j^{**} \cap B_1^*| \leq d \leq p-4$$

contradicting (3.90). By (3.81) for every i with $3 \leq i \leq p$ we have $|B_i \cap B_1^*| = p-2$, which together with (3.79) implies $|B_1^* \setminus B_i| = 1$ ($3 \leq i \leq p$). Thus by (3.91) $B_i \cap B_{j_1}^{**}$ is nonempty for $3 \leq i \leq p$, which again by (3.91) and $B_1^* \subset B_1$ does hold for $i=1$ as well. By (3.88) and (3.89) we thus have

$$\{B_i \cap B_{j_1}^{**} \mid 1 \leq i \leq p, i \neq 2\} = \mathcal{P}_{p-2}(B_{j_1}^{**}). \quad (3.92)$$

Especially $|B_1 \cap B_{j_1}^{**}| = p-2$. By $|B_{j_1}^{**}| = p-1$ there exists $a \in B_{j_1}^{**} \setminus B_1$, i.e. $a \notin B_1 \cap B_{j_1}^{**}$. From (3.92) now follows $a \in B_i \cap B_{j_1}^{**}$ for $3 \leq i \leq p$. By $a \in B_{j_1}^{**} \subset B_2$ we thus see, that $a \in A \setminus B_1$ satisfies (3.82). Thus Theorem 46 follows in subcase B too. $\square$

3.6.3 Open questions

Let $M(m)$ denote the least common multiple of the integers $1, \dots, m$. Let p be an odd prime and A a set with $|A| = M(p-1)$. Then trivially the sequence $B_1 = \dots = B_{p-1} = A$ forms a p -congruence set system on A , which is not p -resolvable on A . We conjecture, that this is the least $|A|$ with this property.

Conjecture 47. *The proposition (3.68) in Theorem 46 may be replaced by $|A| < M(p-1)$.*

Strongly related to this conjecture is the following. Call a p -congruence set system simple, if $p-CSS(A; B_1, \dots, B_n)$ holds, but not $p-CSS(A''; B_1 \cap A'', \dots, B_n \cap A'')$ whenever $A'' \subset A$ with $1 \leq |A''| < |A|$. Examples of simple congruence set systems are the sequence of the $(p-1)$ -element subsets of a p -element set or the congruence set system on a set A with $|A| = M(p-1)$ mentioned above. Theorem 46 says, that the first mentioned p -congruence set system is the only simple one on a set of cardinality at most $(p-3)p$. A solution for the following problem would yield a complete classification of all congruence set systems.

Problem 48. *Determine all simple congruence set systems.*

3.6.4 Proof of Theorem 33

Proof. Let $C \subset \mathbb{F}^n$ be a binary code of length n and covering radius one with $n \equiv -1 \pmod{p}$ for an odd prime p . Assume $\mathbf{x} \in \mathbb{F}^n$ satisfies

$$E_0(\mathbf{x}) = E_1(\mathbf{x}) = \dots = E_{p-2}(\mathbf{x}) = 0. \quad (3.93)$$

We have to show

$$E_{p-1}(\mathbf{x}) \geq (p-2)p - 1. \quad (3.94)$$

It suffices to consider the case $\mathbf{x} = 0$ and we write E_i instead of $E_i(\mathbf{0})$ for $0 \leq i \leq p-1$. By Theorem 40 we have $p-CSS(A; B_1, \dots, B_n)$ for the column sets $B_1, \dots, B_n \subset A$ of the excess matrix $\mathcal{D}$ of C . By (3.93) and Theorem 24 we have $|A| = E_{p-1} \equiv -1 \pmod{p}$ and thus $|A|$ is not divisible by p . An application of Theorem 46 yields $E_{p-1} = |A| > (p-3)p$. Now (3.94) follows by $E_{p-1} \equiv -1 \pmod{p}$. This completes the proof of Theorem 33. $\square$

We remark, that conjecture 47 implies, that the right-hand side of (3.46) in Theorem 33 may be replaced by $M(p-1)$, which grows exponentially with p .

3.7 The general excess bound improved

We now use Theorem 33 to give the following improvement of the general excess bound for binary codes with covering radius one (Theorem 25).

Theorem 49. *If $n \equiv -1 \pmod{p}$ for an odd prime p , then*

$$K(n, 1) \geq \left(1 + \frac{1}{g(n)}\right) \frac{2^n}{n+1}$$

with

$$g(n) = \frac{1}{(p-2)p-1} \binom{n}{p-1} + \sum_{0 \leq i \leq p-2} \frac{1}{i+1} \binom{n}{i}.$$

Proof. Assume $n \equiv -1 \pmod{p}$ for an odd prime p and a positive integer n . Without loss of generality we may assume that n is odd, since for even n Theorem 49 easily follows from Theorem 3. Let $C \subset \mathbb{F}^n$ be a code of length n and covering radius one with $|C| = K(n, 1)$. Assume $p' \leq p$ is the least odd prime satisfying $n \equiv -1 \pmod{p'}$. We may assume $p' \geq 5$, since for $p' = 3$ Theorem 49 already follows from Theorem 6. We use Theorem 33 (with p' instead of p). Let $\mathbf{x} \in \mathbb{F}^n$. If $E_0(\mathbf{x}) = \dots = E_{p'-2}(\mathbf{x}) = 0$ is satisfied, then $E_{p'-1}(\mathbf{x}) \geq (p'-2)p' - 1$. If $E_0(\mathbf{x}) = \dots = E_{p'-2}(\mathbf{x}) = 0$ is not satisfied, let q ($1 \leq q \leq p' - 1 \leq n$) denote the least integer satisfying $E_{q-1}(\mathbf{x}) > 0$. Now q and $n+1$ do not have a common odd prime divisor, since $q < p'$ and p' is assumed to be the least odd prime divisor of $n+1$. Therefore the propositions of Theorem 32 are satisfied. From (3.44), $E_0(\mathbf{x}) = \dots = E_{q-2}(\mathbf{x}) = 0$ and $E_{q-1}(\mathbf{x}) > 0$ now follows $E_{q-1}(\mathbf{x}) \geq q$. Altogether we see, that at least one of the inequalities

$$E_0(\mathbf{x}) \geq 1, E_1(\mathbf{x}) \geq 2, \dots, E_{p'-2}(\mathbf{x}) \geq p' - 1 \text{ and } E_{p'-1}(\mathbf{x}) \geq (p' - 2)p' - 1$$

is satisfied. By $p' \leq p$ and $(p' - 2)p' - 1 \geq p'$ for $p' \geq 5$ this implies, that at least one of the inequalities

$$E_0(\mathbf{x}) \geq 1, E_1(\mathbf{x}) \geq 2, \dots, E_{p-2}(\mathbf{x}) \geq p - 1 \text{ and } E_{p-1}(\mathbf{x}) \geq (p - 2)p - 1$$

must be satisfied. From this we get

$$\frac{1}{(p-2)p-1}E_{p-1}(\mathbf{x}) + \sum_{0 \leq i \leq p-2} \frac{1}{i+1}E_i(\mathbf{x}) \geq 1$$

for each $\mathbf{x} \in \mathbb{F}^n$. Summing over all $\mathbf{x} \in \mathbb{F}^n$ and using

$$\sum_{\mathbf{x} \in \mathbb{F}^n} E_i(\mathbf{x}) = \binom{n}{i} E(\mathbb{F}^n) = \binom{n}{i} ((n+1)|C| - 2^n)$$

for $0 \leq i \leq n$ by (2.8), we get

$$((n+1)|C| - 2^n) \left(\frac{1}{(p-2)p-1} \binom{n}{p-1} + \sum_{0 \leq i \leq p-2} \frac{1}{i+1} \binom{n}{i} \right) \geq 2^n$$

and Theorem 49 follows by $|C| = K(n, 1)$. □

Chapter 4

Bounds for $K(n, R)$

In this chapter we examine bounds for $K(n, R)$, which rely on a covering excess in spheres with radius one. We describe a classical bound of van Wee for this case from 1988 and a slight improvement of this bound by Honkala. We present a short proof of a further improvement of Honkala's bound. Moreover we give an improvement of the van Wee bound in case that $R + 1$ divides n , when both bounds coincide and no general further progress of the van Wee bound is known up to now.

4.1 The bounds of van Wee and Honkala

Let $C \subset \mathbb{F}^n$ be a binary code with covering radius R . We define the set A of deep holes by

$$A = \{\mathbf{x} \in \mathbb{F}^n \mid d(\mathbf{x}, C) = R\}.$$

Trivially we have

$$|A| \geq 2^n - |C|V(n, R - 1). \quad (4.1)$$

The special case $R = 1$ and even n of the following theorem is already contained in Theorem 2. Let $\lceil x \rceil$ denote the smallest integer bigger than or equal to x .

Theorem 50 (van Wee [29]). *Let $C \subset \mathbb{F}^n$ be a binary code with covering radius R . Then for $\mathbf{x} \in A$ we have*

$$E(B(\mathbf{x}, 1)) \geq \epsilon,$$

where ϵ is defined by

$$\epsilon = (R+1) \left\lceil \frac{n+1}{R+1} \right\rceil - (n+1) \leq R. \quad (4.2)$$

Proof. If $\mathbf{c} \in C$, then $|B(\mathbf{c}, R) \cap B(\mathbf{x}, 1)|$ equals the number of words in the sphere $B(\mathbf{x}, 1)$, which were R -covered by $\mathbf{c}$. By $\mathbf{x} \in A$ we have $d(\mathbf{c}, \mathbf{x}) \geq R$. One easily sees, that

$$|B(\mathbf{c}, R) \cap B(\mathbf{x}, 1)| = \begin{cases} R+1 & \text{if } R \leq d(\mathbf{c}, \mathbf{x}) \leq R+1, \\ 0 & \text{if } d(\mathbf{c}, \mathbf{x}) \geq R+2. \end{cases}$$

Now (2.6) applied to $V = B(\mathbf{x}, 1)$ yields

$$\begin{aligned} E(B(\mathbf{x}, 1)) &= \sum_{\mathbf{c} \in C} |B(\mathbf{c}, R) \cap B(\mathbf{x}, 1)| - |B(\mathbf{x}, 1)| \\ &\equiv -(n+1) \pmod{R+1} \\ &\equiv \epsilon \pmod{R+1} \end{aligned}$$

and Theorem 50 follows by $0 \leq \epsilon \leq R$. $\square$

We need another lemma before we can generalize Theorem 3.

Lemma 51. *Let $C \subset \mathbb{F}^n$ be a binary code with covering radius $R \leq n$. Then for $\mathbf{z} \in Z$ we have*

$$|A \cap B(\mathbf{z}, 1)| \leq n - R.$$

Proof. It suffices to consider the case $\mathbf{z} = \mathbf{0}$. By $\mathbf{0} \in Z$ there exist two distinct codewords $\mathbf{c}_1, \mathbf{c}_2 \in C$ with $d(\mathbf{0}, \mathbf{c}_1) \leq R$ and $d(\mathbf{0}, \mathbf{c}_2) \leq R$. We have $\text{supp}(\mathbf{c}_1) \neq \text{supp}(\mathbf{c}_2)$, so without loss $i_0 \in \text{supp}(\mathbf{c}_2) \setminus \text{supp}(\mathbf{c}_1)$ with a suitable i_0 , $1 \leq i_0 \leq n$. If $d(\mathbf{0}, \mathbf{c}_1) \leq R-2$, then $A \cap B(\mathbf{z}, 1) = \emptyset$ and Lemma 51 follows. If $d(\mathbf{0}, \mathbf{c}_1) = R-1$, then clearly $S = \{\mathbf{0}, \mathbf{e}_{i_0}\} \cup \{\mathbf{e}_i \mid i \in \text{supp}(\mathbf{c}_1)\} \subset B(\mathbf{0}, 1) \setminus (A \cap B(\mathbf{0}, 1))$ and Lemma 51 follows by $|S| = R+1$ and $|B(\mathbf{0}, 1)| = n+1$. If $d(\mathbf{0}, \mathbf{c}_1) = R$, then $S = \{\mathbf{e}_{i_0}\} \cup \{\mathbf{e}_i \mid i \in \text{supp}(\mathbf{c}_1)\} \subset B(\mathbf{0}, 1) \setminus (A \cap B(\mathbf{0}, 1))$ and Lemma 51 follows again. $\square$

The following theorem of van Wee [29] generalizes Theorem 3 to arbitrary covering radius.

Theorem 52 (van Wee [29]). *If $1 \leq R < n$, then*

$$K(n, R) \geq \frac{(n - R + \epsilon)2^n}{(n - R)V(n, R) + \epsilon V(n, R - 1)},$$

where ϵ is defined in (4.2).

Proof. Let $C \subset \mathbb{F}^n$ be a binary code with covering radius R and $|C| = K(n, R)$. We get

$$\begin{aligned} \epsilon|A| &= \sum_{\mathbf{x} \in A} \epsilon \\ &\leq \sum_{\mathbf{x} \in A} E(B(\mathbf{x}, 1)) \quad \text{by Theorem 50} \\ &= \sum_{\mathbf{x} \in A} \sum_{\mathbf{z} \in B(\mathbf{x}, 1)} E(\mathbf{z}) \\ &= \sum_{\mathbf{z} \in Z} E(\mathbf{z}) \sum_{\mathbf{x} \in A \cap B(\mathbf{z}, 1)} 1 \\ &\leq (n - R)E(\mathbb{F}^n) \quad \text{by Lemma 51} \\ &= (n - R)(|C|V(n, R) - 2^n) \quad \text{by (2.7)} \end{aligned}$$

and Theorem 52 follows by (4.1) and $|C| = K(n, R)$. $\square$

Theorem 52 was further improved by Honkala [13], [14].

Theorem 53 (Honkala [13], [14]). *If $R \geq 2$, $n \geq 2R + 1$ and ϵ (defined in (4.2)) satisfies*

$$\epsilon \leq R - 1,$$

then

$$K(n, R) \geq \frac{(\rho + \epsilon)2^n}{\rho V(n, R) + \epsilon V(n, R - 1)},$$

where

$$\rho = \begin{cases} n - 3 + \frac{2}{n} & \text{if } R = 2, \\ n - R - 1 & \text{if } R \geq 3. \end{cases} \quad (4.3)$$

The proof of Theorem 53 works by dividing Z into several classes, use improved estimations for some of them and then average in a suitable manner. We omit the proof, since it is somewhat complicated and in the next section we present a quite short proof of a further improvement of Theorem 53.

4.2 An improvement of Honkala's bound

In this section we prove the following theorem.

Theorem 54. *If $R \geq 1$, $n \geq 2R + 1$ and ϵ is defined in (4.2), then*

$$K(n, R) \geq \frac{(n - 2R + 2\epsilon)2^n}{(n - 2R + \epsilon)V(n, R) + \epsilon V(n, R - 1)}.$$

Especially the extra summand $2/n$ in (4.3) in case of $R = 2$ may be omitted. Note, that the restriction $n \geq 2R + 1$ is not essential, since all values of $K(n, R)$ are known, when $n < 2R + 1$, see [4].

As a corollary we get the following improvements for K ris table [19].

Corollary 55.

$$\begin{aligned} K(19, 2) &\geq 2898 && (2897 [14]) \\ K(25, 2) &\geq 107218 && (107203 [14]) \\ K(28, 2) &\geq 683980 && (683972 [10]) \\ K(31, 2) &\geq 4464944 && (4464613 [14]) \\ K(32, 4) &\geq 110467 && (110215 [14]). \end{aligned}$$

Proof of Theorem 54. Let $C \subset \mathbb{F}^n$ be a binary code with covering radius R and $|C| = K(n, R)$. We set

$$\begin{aligned} A_i &= \{\mathbf{x} \in A \mid E(B(\mathbf{x}, 1)) = i\} && (\epsilon \leq i < R), \\ A_R &= \{\mathbf{x} \in A \mid E(B(\mathbf{x}, 1)) \geq R\}. \end{aligned}$$

Moreover for $\epsilon \leq i < R$ let

$$\begin{aligned} Z^{(i)} &= \{\mathbf{z} \in Z \mid A_i \cap B(\mathbf{z}, 1) \neq \emptyset, A_j \cap B(\mathbf{z}, 1) = \emptyset \text{ for all } \epsilon \leq j < i\}, \\ Z^{(R)} &= Z \setminus \bigcup_{\epsilon \leq i < R} Z^{(i)}. \end{aligned} \tag{4.4}$$

By definition we have ($\epsilon \leq i < R$)

$$A_i \cap B(\mathbf{z}, 1) = \emptyset \quad \text{if } \mathbf{z} \in Z^{(j)} \text{ with } i < j \leq R. \tag{4.5}$$

We now improve on Lemma 51.

Lemma 56. *If $\epsilon \leq i \leq R$ and $\mathbf{z} \in Z^{(i)}$, then*

$$|A \cap B(\mathbf{z}, 1)| \leq n - 2R + i.$$

Proof. If $i = R$, then Lemma 56 follows from Lemma 51, so we may assume $\epsilon \leq i < R$. By (4.4) we may choose $\mathbf{x} \in A_i \cap B(\mathbf{z}, 1)$. By $\mathbf{z} \in Z$ there exist $\mathbf{c}_1, \mathbf{c}_2 \in C$ with $\mathbf{c}_1 \neq \mathbf{c}_2$ and $d(\mathbf{c}_1, \mathbf{z}), d(\mathbf{c}_2, \mathbf{z}) \leq R$. By $\mathbf{x} \in A$ and $d(\mathbf{z}, \mathbf{x}) \leq 1$ we have $d(\mathbf{c}_1, \mathbf{x}), d(\mathbf{c}_2, \mathbf{x}) \in \{R, R+1\}$. We only consider the case $d(\mathbf{c}_1, \mathbf{x}) = d(\mathbf{c}_2, \mathbf{x}) = R+1$ (implying $\mathbf{x} \neq \mathbf{z}$), since the other cases are working quite similar. Assume $\mathbf{x} = (x_1, \dots, x_n)$, $\mathbf{c}_i = (c_{i1}, \dots, c_{in})$ for $i \in \{1, 2\}$ and

$$B_i = \{1 \leq j \leq n \mid c_{ij} \neq x_j\}, \quad i \in \{1, 2\}.$$

Clearly $\mathbf{x} + \mathbf{e}_k$ is R -covered both by $\mathbf{c}_1$ as well as by $\mathbf{c}_2$ whenever $k \in B_1 \cap B_2$, implying $|B_1 \cap B_2| \leq \sum_{k \in B_1 \cap B_2} E(\mathbf{x} + \mathbf{e}_k) \leq E(B(\mathbf{x}, 1)) = i$ by $\mathbf{x} \in A_i$. Therefore we have

$$|B_1 \cup B_2| = |B_1| + |B_2| - |B_1 \cap B_2| = 2R + 2 - |B_1 \cap B_2| \geq 2R - i + 2.$$

Fix $j_0 \in \{1, \dots, n\}$ with $\mathbf{z} = \mathbf{x} + \mathbf{e}_{j_0}$. Now for every $k \in (B_1 \cup B_2) \setminus \{j_0\}$ we either have $d(\mathbf{z} + \mathbf{e}_k, \mathbf{c}_1) < R$ or $d(\mathbf{z} + \mathbf{e}_k, \mathbf{c}_2) < R$, which means $\mathbf{z} + \mathbf{e}_k \notin A$ for $k \in (B_1 \cup B_2) \setminus \{j_0\}$. This implies

$$\begin{aligned} |A \cap B(\mathbf{z}, 1)| &\leq |B(\mathbf{z}, 1)| - |(B_1 \cup B_2) \setminus \{j_0\}| \\ &= n + 1 - |(B_1 \cup B_2) \setminus \{j_0\}| \\ &\leq n - 2R + i, \end{aligned}$$

completing the proof of Lemma 56. □

We now are able to prove Theorem 54. We may assume $\epsilon > 0$, since the

case $\epsilon = 0$ reduces to the sphere-covering bound (2.4). We have

$$\begin{aligned}
|A| &= \sum_{\epsilon \leq i \leq R} |A_i| \quad \text{by Theorem 50} \\
&\leq \sum_{\epsilon \leq i \leq R} \frac{1}{i} \sum_{\mathbf{x} \in A_i} E(B(\mathbf{x}, 1)) \\
&= \sum_{\epsilon \leq i \leq R} \frac{1}{i} \sum_{\mathbf{z} \in Z} E(\mathbf{z}) \sum_{\mathbf{x} \in A_i \cap B(\mathbf{z}, 1)} 1 \\
&= \sum_{\epsilon \leq j \leq R} \sum_{\mathbf{z} \in Z^{(j)}} E(\mathbf{z}) \sum_{\epsilon \leq i \leq R} \frac{1}{i} \sum_{\mathbf{x} \in A_i \cap B(\mathbf{z}, 1)} 1 \\
&= \sum_{\epsilon \leq j \leq R} \sum_{\mathbf{z} \in Z^{(j)}} E(\mathbf{z}) \sum_{j \leq i \leq R} \frac{1}{i} \sum_{\mathbf{x} \in A_i \cap B(\mathbf{z}, 1)} 1 \quad \text{by (4.5)} \\
&\leq \sum_{\epsilon \leq j \leq R} \frac{1}{j} \sum_{\mathbf{z} \in Z^{(j)}} E(\mathbf{z}) \sum_{\mathbf{x} \in A \cap B(\mathbf{z}, 1)} 1 \\
&\leq \sum_{\epsilon \leq j \leq R} \frac{n - 2R + j}{j} \sum_{\mathbf{z} \in Z^{(j)}} E(\mathbf{z}) \quad \text{by Lemma 56} \\
&\leq \frac{n - 2R + \epsilon}{\epsilon} \sum_{\mathbf{z} \in Z} E(\mathbf{z}) \quad \text{by } n \geq 2R + 1 \\
&= \frac{n - 2R + \epsilon}{\epsilon} E(\mathbb{F}^n)
\end{aligned}$$

and Theorem 54 follows using (2.7), (4.1) and $|C| = K(n, R)$. $\square$

4.3 The case $\epsilon = R$

Of special interest is the case $\epsilon = R$ (which means, that $R + 1$ divides n), since then van Wee's bound (Theorem 52) and the new bound (Theorem 54) coincide and the propositions of Honkala's bound (Theorem 53) are not satisfied. Indeed (as far as the author is aware) in that case no general improvement of the van Wee bound is known. In this section we present such an improvement.

Theorem 57. *If $2 \leq R < n$, $R + 1$ divides n and*

$$M = \frac{\binom{R}{2} + R - 1}{V(n, 2) - 1 - (R - 2) \left(n - \frac{R-3}{2}\right)} \leq \frac{1}{n} \quad (4.6)$$

holds, then

$$K(n, R) \geq \frac{(n - \tau)2^n}{(n - R - \tau)V(n, R) + RV(n, R - 1)}$$

with

$$\tau = (n - R) \frac{M}{M + 1}.$$

It is easy to see, that the denominator of the fraction in (4.6) is strictly positive for $2 \leq R < n$. Thus $M > 0$, implying $\tau > 0$, which means, that Theorem 57 indeed improves on the van Wee bound (Theorem 52). Moreover condition (4.6) is usually satisfied, for instance if R is fixed and n is large enough.

For the rest of the section we always assume $R \geq 2$.

Corollary 58.

$$\begin{aligned} K(21, 2) &\geq 9900 && (9893 [29]) \\ K(24, 2) &\geq 60381 && (60350 [29]) \\ K(33, 2) &\geq 16210701 && (16207424 [29]) \\ K(20, 3) &\geq 892 && (889 [29]) \\ K(32, 3) &\geq 855613 && (854890 [29]) \\ K(30, 4) &\geq 38067 && (37973 [29]). \end{aligned}$$

Let $C \subset \mathbb{F}^n$ be a binary code with covering radius R and $|C| = K(n, R)$. For the proof of Theorem 57 we use the notations of the sections 4.1 and 4.2. Moreover we set

$$\begin{aligned} Z_{(1)} &= \{\mathbf{z} \in Z \mid |A \cap B(\mathbf{z}, 1)| = n - R\}, \\ Z_{(2)} &= \{\mathbf{z} \in Z \mid A \cap B(\mathbf{z}, 1) = \emptyset\}, \\ Z_{(3)} &= Z \setminus (Z_{(1)} \cup Z_{(2)}). \end{aligned}$$

We have

$$Z_{(1)} \cap B(C, R - 2) = \emptyset, \tag{4.7}$$

where $B(C, i)$, $0 \leq i \leq n$ is defined by

$$B(C, i) = \bigcup_{\mathbf{c} \in C} B(\mathbf{c}, i) \quad (0 \leq i \leq n).$$

If not, assume $\mathbf{z} \in Z_{(1)} \cap B(C, R-2)$. Then there exists $\mathbf{c} \in C$ with $d(\mathbf{z}, \mathbf{c}) \leq R-2$. This implies $d(\mathbf{x}, \mathbf{c}) \leq R-1$ for every $\mathbf{x} \in B(\mathbf{z}, 1)$ and thus $A \cap B(\mathbf{z}, 1) = \emptyset$, contradicting $|A \cap B(\mathbf{z}, 1)| = n - R > 0$ for $\mathbf{z} \in Z_{(1)}$.

For the proof of Theorem 57 we need three more lemmas.

Lemma 59. *If $\mathbf{y}, \mathbf{c} \in \mathbb{F}^n$ with $d(\mathbf{y}, \mathbf{c}) \leq R-2$, then*

$$|B(\mathbf{y}, 2) \cap B(\mathbf{c}, R-2)| \geq 1 + (R-2) \left(n - \frac{R-3}{2} \right). \quad (4.8)$$

Proof. It suffices to consider the case $d(\mathbf{y}, \mathbf{c}) = R-2$, since it is well-known, that $|B(\mathbf{x}_1, r_1) \cap B(\mathbf{x}_2, r_2)| \geq |B(\mathbf{x}_3, r_1) \cap B(\mathbf{x}_4, r_2)|$ whenever $\mathbf{x}_1, \mathbf{x}_2, \mathbf{x}_3, \mathbf{x}_4 \in \mathbb{F}^n$ satisfy $d(\mathbf{x}_1, \mathbf{x}_2) \leq d(\mathbf{x}_3, \mathbf{x}_4)$ (van Wee [29], see also [4] Theorem 2.4.8). Assume $\mathbf{x} \in B(\mathbf{y}, 2) \cap B(\mathbf{c}, R-2)$. Then either $\mathbf{x} = \mathbf{y}$ or $\mathbf{x}$ is obtained from $\mathbf{y}$ by changing $\mathbf{y}$ in one of the $R-2$ coordinates, where $\mathbf{y}$ differs from $\mathbf{c}$ and in zero or one of the $n-1$ remaining coordinates. Now every $\mathbf{x} \in B(\mathbf{y}, 2) \cap B(\mathbf{c}, R-2)$ obtained from $\mathbf{y}$ by changing $\mathbf{y}$ in two of the $R-2$ coordinates, where $\mathbf{c}$ and $\mathbf{y}$ differ, is counted twice. Altogether

$$|B(\mathbf{y}, 2) \cap B(\mathbf{c}, R-2)| = 1 + (R-2)n - \binom{R-2}{2}$$

and (4.8) follows. $\square$

A part of the next lemma already appeared in Habsieger, Plagne [10] (Lemma 6).

Lemma 60. *For every $\mathbf{z} \in Z_{(1)}$ there exists a set $D_{\mathbf{z}}$ satisfying*

- a) $D_{\mathbf{z}} \subset (B(\mathbf{z}, 2) \setminus \{\mathbf{z}\}) \cap B(C, R-2) \cap Z_{(2)},$
- b) $|D_{\mathbf{z}}| = \binom{R}{2} + R - 1,$
- c) $E(\mathbf{y}) \geq E(\mathbf{z}) \quad \text{for every } \mathbf{y} \in D_{\mathbf{z}}.$

Proof. It suffices to assume $\mathbf{z} = \mathbf{0}$. By $\mathbf{0} \in Z_{(1)} \subset Z$ there exist $\mathbf{c}_1, \mathbf{c}_2 \in C$ with $\mathbf{c}_1 \neq \mathbf{c}_2$, $d(\mathbf{c}_1, \mathbf{0}) \leq R$ and $d(\mathbf{c}_2, \mathbf{0}) \leq R$. We have

$$w(\mathbf{c}_1 \cap \mathbf{c}_2) \leq R-1, \quad (4.9)$$

since $w(\mathbf{c}_1 \cap \mathbf{c}_2) \geq R$ would imply $\mathbf{c}_1 = \mathbf{c}_2$. We set

$$D_j(\mathbf{u}) = \{\mathbf{x} \in \mathbb{F}^n \mid w(\mathbf{x}) = j, \mathbf{x} \subset \mathbf{u}\}$$

for $\mathbf{u} \in \mathbb{F}^n$, $j \in \{1, 2\}$. Apparently

$$|D_j(\mathbf{u})| = \binom{w(\mathbf{u})}{j}, \quad \mathbf{u} \in \mathbb{F}^n, \quad j \in \{1, 2\}. \quad (4.10)$$

From (4.7) follows

$$d(\mathbf{c}_i, \mathbf{0}) \in \{R-1, R\}, \quad i \in \{1, 2\}.$$

Therefore it suffices to consider the following three cases.

Case I. $d(\mathbf{c}_1, \mathbf{0}) = d(\mathbf{c}_2, \mathbf{0}) = R$.

We have $d(\mathbf{x}, \mathbf{c}_i) = R-1$ for $\mathbf{x} \in D_1(\mathbf{c}_i)$, $i \in \{1, 2\}$, implying $A \cap (D_1(\mathbf{c}_1) \cup D_1(\mathbf{c}_2)) = \emptyset$. Thus $D_1(\mathbf{c}_1) \cup D_1(\mathbf{c}_2) \subset B(\mathbf{0}, 1) \setminus (A \cap B(\mathbf{0}, 1))$ implying

$$\begin{aligned} w(\mathbf{c}_1 \cup \mathbf{c}_2) &= |D_1(\mathbf{c}_1 \cup \mathbf{c}_2)| \quad \text{by (4.10)} \\ &= |D_1(\mathbf{c}_1) \cup D_1(\mathbf{c}_2)| \\ &\leq |B(\mathbf{0}, 1)| - |A \cap B(\mathbf{0}, 1)| \\ &= n+1 - (n-R) \quad \text{by } \mathbf{0} \in Z_{(1)} \\ &= R+1. \end{aligned}$$

From this follows

$$\begin{aligned} w(\mathbf{c}_1 \cap \mathbf{c}_2) &= |D_1(\mathbf{c}_1 \cap \mathbf{c}_2)| \quad \text{by (4.10)} \\ &= |D_1(\mathbf{c}_1) \cap D_1(\mathbf{c}_2)| \\ &= |D_1(\mathbf{c}_1)| + |D_1(\mathbf{c}_2)| - |D_1(\mathbf{c}_1) \cup D_1(\mathbf{c}_2)| \\ &\geq 2R - (R+1) \quad \text{by (4.10)} \\ &= R-1. \end{aligned}$$

By (4.9) this implies

$$w(\mathbf{c}_1 \cap \mathbf{c}_2) = R-1 \quad (4.11)$$

and thus

$$\begin{aligned} d(\mathbf{c}_1, \mathbf{c}_2) &= w(\mathbf{c}_1 \cup \mathbf{c}_2) - w(\mathbf{c}_1 \cap \mathbf{c}_2) \\ &\leq R+1 - (R-1) \\ &= 2. \end{aligned} \quad (4.12)$$

We now set

$$D_0 = D_2(\mathbf{c}_1) \cup D_2(\mathbf{c}_2) \subset (B(\mathbf{0}, 2) \setminus \{\mathbf{0}\}) \cap B(\mathbf{0}, R-2).$$

To prove a) it remains to show $D_0 \subset Z_{(2)}$. Assume $\mathbf{y} \in D_0$, say $\mathbf{y} \in D_2(\mathbf{c}_1)$. Then $d(\mathbf{y}, \mathbf{c}_1) = R - 2$, implying $d(\mathbf{y}, \mathbf{c}_2) \leq d(\mathbf{y}, \mathbf{c}_1) + d(\mathbf{c}_1, \mathbf{c}_2) \leq R$ by (4.12) and thus $\mathbf{y} \in Z$. Again by $d(\mathbf{y}, \mathbf{c}_1) = R - 2$ we find $d(\mathbf{x}, \mathbf{c}_1) \leq R - 1$ for every $\mathbf{x} \in B(\mathbf{y}, 1)$. Thus $A \cap B(\mathbf{y}, 1) = \emptyset$ implying $\mathbf{y} \in Z_{(2)}$ and $D_0 \subset Z_{(2)}$ follows. To show b) we have

$$\begin{aligned}
|D_0| &= |D_2(\mathbf{c}_1)| + |D_2(\mathbf{c}_2)| - |D_2(\mathbf{c}_1) \cap D_2(\mathbf{c}_2)| \\
&= |D_2(\mathbf{c}_1)| + |D_2(\mathbf{c}_2)| - |D_2(\mathbf{c}_1 \cap \mathbf{c}_2)| \\
&= \binom{R}{2} + \binom{R}{2} - \binom{R-1}{2} \quad \text{by (4.10) and (4.11)} \\
&= \binom{R}{2} + R - 1.
\end{aligned}$$

Case II. $d(\mathbf{c}_1, \mathbf{0}) = R - 1$, $d(\mathbf{c}_2, \mathbf{0}) = R$.

Similar like in case I we find $\mathbf{c}_1 \subset \mathbf{c}_2$, $w(\mathbf{c}_1 \cap \mathbf{c}_2) = R - 1$ and $w(\mathbf{c}_1 \cup \mathbf{c}_2) = R$. We set

$$D_0 = D_1(\mathbf{c}_1) \cup D_2(\mathbf{c}_2)$$

and a), b) follow similar like in case I. Moreover $d(\mathbf{c}_1, \mathbf{c}_2) = 1$, so that (4.12) is satisfied.

Case III. $d(\mathbf{c}_1, \mathbf{0}) = d(\mathbf{c}_2, \mathbf{0}) = R - 1$.

Here we find $w(\mathbf{c}_1 \cap \mathbf{c}_2) = R - 2$, $w(\mathbf{c}_1 \cup \mathbf{c}_2) = R$ and we set

$$D_0 = D_1(\mathbf{c}_1) \cup D_1(\mathbf{c}_2) \cup D_2(\mathbf{c}_1) \cup D_2(\mathbf{c}_2).$$

Again a) and b) follow and (4.12) is satisfied. This completes the proof of Lemma 60 a) and b).

It remains to show c). Assume $\mathbf{y} \in D_0$. From (4.12) (valid in all cases) follows

$$d(\mathbf{c}, \mathbf{c}') \leq 2 \tag{4.13}$$

whenever $\mathbf{c}, \mathbf{c}' \in C$ with $d(\mathbf{c}, \mathbf{0}) \leq R$ and $d(\mathbf{c}', \mathbf{0}) \leq R$. Now c) follows, if we are able to show $d(\mathbf{c}, \mathbf{y}) \leq R$ for every $\mathbf{c} \in C$ with $d(\mathbf{c}, \mathbf{0}) \leq R$. Assume $\mathbf{c} \in C$ with $d(\mathbf{c}, \mathbf{0}) \leq R$. By $\mathbf{y} \in D_0 \subset B(C, R - 2)$ there exists $\mathbf{c}' \in C$ with $d(\mathbf{c}', \mathbf{y}) \leq R - 2$. By $\mathbf{y} \in B(\mathbf{0}, 2)$ we have $d(\mathbf{c}', \mathbf{0}) \leq d(\mathbf{c}', \mathbf{y}) + d(\mathbf{y}, \mathbf{0}) \leq R - 2 + 2 = R$. Thus $d(\mathbf{c}, \mathbf{c}') \leq 2$ by (4.13) and we get $d(\mathbf{c}, \mathbf{y}) \leq d(\mathbf{c}, \mathbf{c}') + d(\mathbf{c}', \mathbf{y}) \leq 2 + R - 2 = R$ by (4.13). This completes the proof of Lemma 60. $\square$

Lemma 61.

$$\begin{aligned} & \left(\binom{R}{2} + R - 1 \right) \sum_{\mathbf{z} \in Z_{(1)}} E(\mathbf{z}) \\ & \leq \left(V(n, 2) - 1 - (R - 2) \left(n - \frac{R - 3}{2} \right) \right) \sum_{\mathbf{z} \in Z_{(2)}} E(\mathbf{z}). \end{aligned}$$

Proof. For $\mathbf{z} \in Z_{(1)}$ let $D_{\mathbf{z}}$ denote a set satisfying a), b) and c) from Lemma 60. For $i \geq 1$ we set

$$\begin{aligned} D &= \bigcup_{\mathbf{z} \in Z_{(1)}} D_{\mathbf{z}}, \\ Z_{(1),i} &= \{\mathbf{z} \in Z_{(1)} \mid E(\mathbf{z}) = i\}, \\ D_i &= \{\mathbf{y} \in D \mid E(\mathbf{y}) = i\}. \end{aligned}$$

Moreover we use the abbreviation $(i, j \geq 1)$

$$a_{ij} = \sum_{\mathbf{z} \in Z_{(1),i}} \sum_{\mathbf{y} \in D_j \cap D_{\mathbf{z}}} 1.$$

Assume $\mathbf{z} \in Z_{(1)}$ and $\mathbf{y} \in D_{\mathbf{z}}$. Then by Lemma 60 a) $\mathbf{y} \in D_{\mathbf{z}} \subset B(\mathbf{z}, 2) \cap B(C, R - 2)$. Thus $\mathbf{z} \in B(\mathbf{y}, 2)$ and $d(\mathbf{y}, \mathbf{c}_{\mathbf{y}}) \leq R - 2$ for a suitable $\mathbf{c}_{\mathbf{y}} \in C$. Now by $\mathbf{z} \in Z_{(1)}$ and (4.7) we get $\mathbf{z} \notin B(\mathbf{c}_{\mathbf{y}}, R - 2)$. Fixing $\mathbf{y} \in D$ we thus have

$$\begin{aligned} |\{\mathbf{z} \in Z_{(1)} \mid \mathbf{y} \in D_{\mathbf{z}}\}| &\leq |B(\mathbf{y}, 2) \setminus (B(\mathbf{y}, 2) \cap B(\mathbf{c}_{\mathbf{y}}, R - 2))| \quad (4.14) \\ &\leq V(n, 2) - 1 - (R - 2) \left(n - \frac{R - 3}{2} \right) \end{aligned}$$

by Lemma 59. Fixing $i \geq 1$ we now get

$$\begin{aligned}
& \left(\binom{R}{2} + R - 1 \right) |Z_{(1),i}| - \sum_{j>i} a_{ij} \\
&= \sum_{\mathbf{z} \in Z_{(1),i}} \sum_{\mathbf{y} \in D_{\mathbf{z}}} 1 - \sum_{j>i} \sum_{\mathbf{z} \in Z_{(1),i}} \sum_{\mathbf{y} \in D_j \cap D_{\mathbf{z}}} 1 \quad \text{by Lemma 60 b)} \\
&= \sum_{\mathbf{z} \in Z_{(1),i}} \sum_{\mathbf{y} \in D_i \cap D_{\mathbf{z}}} 1 \quad \text{by Lemma 60 c)} \\
&= \sum_{\mathbf{y} \in D_i} \sum_{\substack{\mathbf{z} \in Z_{(1),i} \\ \mathbf{y} \in D_{\mathbf{z}}}} 1 \\
&= \sum_{\mathbf{y} \in D_i} \sum_{\substack{\mathbf{z} \in Z_{(1)} \\ \mathbf{y} \in D_{\mathbf{z}}}} 1 - \sum_{j<i} \sum_{\mathbf{y} \in D_i} \sum_{\substack{\mathbf{z} \in Z_{(1),j} \\ \mathbf{y} \in D_{\mathbf{z}}}} 1 \quad \text{by Lemma 60 c)} \\
&\leq \sum_{\mathbf{y} \in D_i} \left(V(n, 2) - 1 - (R - 2) \left(n - \frac{R - 3}{2} \right) \right) - \sum_{j<i} \sum_{\mathbf{z} \in Z_{(1),j}} \sum_{\mathbf{y} \in D_i \cap D_{\mathbf{z}}} 1 \\
&\quad \text{by (4.14)} \\
&= \left(V(n, 2) - 1 - (R - 2) \left(n - \frac{R - 3}{2} \right) \right) |D_i| - \sum_{j<i} a_{ji}.
\end{aligned}$$

Multiplying with i and sum over all $i \geq 1$ we get

$$\begin{aligned}
& \left(\binom{R}{2} + R - 1 \right) \sum_{\mathbf{z} \in Z_{(1)}} E(\mathbf{z}) \\
&= \left(\binom{R}{2} + R - 1 \right) \sum_{i \geq 1} i |Z_{(1),i}| \\
&\leq \left(V(n, 2) - 1 - (R - 2) \left(n - \frac{R - 3}{2} \right) \right) \sum_{i \geq 1} i |D_i| \\
&\quad + \sum_{i \geq 1} i \sum_{j>i} a_{ij} - \sum_{i \geq 1} i \sum_{j<i} a_{ji} \\
&= \left(V(n, 2) - 1 - (R - 2) \left(n - \frac{R - 3}{2} \right) \right) \sum_{\mathbf{y} \in D} E(\mathbf{y}) - \sum_{1 \leq i < j} (j - i) a_{ij} \\
&\leq \left(V(n, 2) - 1 - (R - 2) \left(n - \frac{R - 3}{2} \right) \right) \sum_{\mathbf{z} \in Z_{(2)}} E(\mathbf{z}),
\end{aligned}$$

since $D \subset Z_{(2)}$ by Lemma 60 a). $\square$

Proof of Theorem 57. Let $C \subset \mathbb{F}^n$ be a binary code with covering radius R and $|C| = K(n, R)$. Assume

$$\sum_{\mathbf{z} \in Z_{(i)}} E(\mathbf{z}) = x_i \sum_{\mathbf{z} \in Z} E(\mathbf{z}) \quad (1 \leq i \leq 3),$$

with suitable nonnegative real numbers x_1, x_2, x_3 . By $R < n$ we see, that the sets $Z_{(1)}$, $Z_{(2)}$ and $Z_{(3)}$ are pairwise disjoint and satisfy $Z_{(1)} \cup Z_{(2)} \cup Z_{(3)} = Z$, implying

$$x_1 + x_2 + x_3 = 1. \quad (4.15)$$

Moreover by Lemma 51

$$|A \cap B(\mathbf{z}, 1)| \leq n - R - 1 \quad \text{if } \mathbf{z} \in Z_{(3)}. \quad (4.16)$$

Remember that M is defined in (4.6). By Lemma 61 we have

$$x_2 \geq Mx_1, \quad (4.17)$$

which by (4.15) implies $x_1 \leq 1 - x_2 \leq 1 - Mx_1$. Thus

$$x_1 \leq \frac{1}{M+1}. \quad (4.18)$$

Since $\epsilon = R$ if $R+1$ divides n , we find

$$\begin{aligned} R|A| &\leq \sum_{\mathbf{x} \in A} E(B(\mathbf{x}, 1)) \quad \text{by Theorem 50} \\ &= \sum_{\mathbf{z} \in Z} E(\mathbf{z}) \sum_{\mathbf{x} \in A \cap B(\mathbf{z}, 1)} 1 \\ &\leq (n-R) \sum_{\mathbf{z} \in Z_{(1)}} E(\mathbf{z}) + (n-R-1) \sum_{\mathbf{z} \in Z_{(3)}} E(\mathbf{z}) \\ &\quad (\text{by Lemma 51 and (4.16)}) \\ &= (n-R) \sum_{\mathbf{z} \in Z} E(\mathbf{z}) - (n-R) \sum_{\mathbf{z} \in Z_{(2)}} E(\mathbf{z}) - \sum_{\mathbf{z} \in Z_{(3)}} E(\mathbf{z}) \\ &= (n-R-(n-R)x_2-x_3) \sum_{\mathbf{z} \in Z} E(\mathbf{z}). \end{aligned} \quad (4.19)$$

We have

$$\begin{aligned}
(n-R)x_2 + x_3 &= 1 - x_1 + (n-R-1)x_2 && \text{by (4.15)} \\
&\geq 1 - (1 - (n-R-1)M)x_1 && \text{by (4.17)} \\
&\geq 1 - (1 - (n-R-1)M)\frac{1}{M+1} && \text{by (4.6) and (4.18)} \\
&= (n-R)\frac{M}{M+1} \\
&= \tau.
\end{aligned}$$

This inserted in (4.19) yields

$$R|A| \leq (n-R-\tau) \sum_{\mathbf{z} \in Z} E(\mathbf{z}) = (n-R-\tau)E(\mathbb{F}^n)$$

and Theorem 57 follows by (2.7), (4.1) and $|C| = K(n, R)$. $\square$

Appendix A

Further results

Here we shortly resume further work, which was done by various authors using the method of covering excess. Much of this work generalizes the set $\mathbb{F} = \{0, 1\}$ to an alphabet consisting of q letters. Then $A_q(n, d)$ denotes the maximal cardinality of a q -ary code with minimum Hamming distance d and $K_q(n, R)$ the minimal cardinality of a q -ary code of length n and covering radius R . Of special interest is the ternary case $q = 3$ due to its connection to football pools. So for instance the determination of $K_3(n, 1)$ is widely known as the football pool problem (for n matches): try to forecast the outcome of n football matches, each having three possible results (including a tie). You get a second price, if one of your forecasts has $n - 1$ correct results. Then $K_3(n, 1)$ equals the least number of forecasts to place, which, no matter of the outcome of the matches, guarantee at least a second price.

The method of covering excess is often combined with the use of linear inequalities on the quantities $A_i(\mathbf{x})$ defined in (3.27).

A.1 Notes to Chapter 2 and 3

Generalizations of the binary Hamming codes (Theorem 1) to the q -ary case can be found in every textbook on coding theory, see for instance Cohen et al. [4], or in an early paper of Shapiro and Slotnick [28].

An improvement of the bound $K(n, 1) \geq 2^n/n$ for even n (Theorem 3) is due to Habsieger [9], if additionally n is divisible by 5, but not divisible by 3. In all other cases (apart from a few sporadic values of n) Theorem 3 gives the currently best known lower bound on $K(n, 1)$ if n is even. A generalization

of Theorem 3 to lower-bound $K_q(n, 1)$ if q and n are even, can be found in van Wee [30].

A analogous approach to the theory of covering excess can be used to upper-bound $A_q(n, d)$, see van Wee [30].

A covering excess in spheres of radius two corresponding to Theorem 5 in the ternary case $q = 3$ and $R = 1$ (occurring if $n \equiv 0 \pmod{3}$ or $n \equiv 2 \pmod{3}$) is studied in Habsieger [8] and van Wee [30].

A.2 Notes to Chapter 4

Generalizations of Theorem 52 to the q -ary case can be found in Chen, Honkala [3] and van Wee [30]. Lower-bounding $K(n, R)$ with the use of a covering excess in spheres with radius two is studied by Honkala [14] and in the q -ary case by Chen, Honkala [3]. Lower bounds for $K_q(n, R)$ in the ternary case $q = 3$ can also be found in Habsieger [8].

A covering excess in spheres of radius two in the binary case, based on the nonexistence of certain pair covering designs, was already investigated in the early paper of Johnson [18]. The method was later rediscovered by Zhang [32]. The corresponding ternary case $q = 3$ is studied in an unpublished paper from Lo and Zhen [24].

Many of the currently best known lower bounds for $K(n, R)$ result from a similar covering excess in a sphere of radius three, see the papers Li, Chen [21] and Lo, Zhang [23].

Application of the method of covering excess to binary/ternary mixed codes are considered in van Wee [30] and especially in van Lint jr., van Wee [22].

Appendix B

The new bounds

Here we collect all new bounds from this thesis.

Theorem 14 (Haas [7]). If $n \geq 11$ and $n \equiv 5 \pmod{6}$, then

$$K(n, 1) \geq \left(1 + \frac{8}{2n^2 - 10n + 19}\right) \frac{2^n}{n+1}.$$

Corollary.

$$\begin{aligned} K(17, 1) &\geq 7419 \quad (7414 [1]), \\ K(23, 1) &\geq 352827 \quad (352736 [1]), \\ K(29, 1) &\geq 17997161 \quad (17994966 [1]). \end{aligned}$$

Theorem 49. If $n \equiv -1 \pmod{p}$ for an odd prime p , then

$$K(n, 1) \geq \left(1 + \frac{1}{g(n)}\right) \frac{2^n}{n+1}$$

with

$$g(n) = \frac{1}{(p-2)p-1} \binom{n}{p-1} + \sum_{0 \leq i \leq p-2} \frac{1}{i+1} \binom{n}{i}.$$

Theorem 54. If $R \geq 1$ and $n \geq 2R+1$, then

$$K(n, R) \geq \frac{(n-2R+2\epsilon)2^n}{(n-2R+\epsilon)V(n, R) + \epsilon V(n, R-1)}$$

with

$$\epsilon = (R + 1) \left\lceil \frac{n + 1}{R + 1} \right\rceil - (n + 1) \leq R.$$

Corollary.

$$\begin{aligned} K(19, 2) &\geq 2898 && (2897 [14]) \\ K(25, 2) &\geq 107218 && (107203 [14]) \\ K(28, 2) &\geq 683980 && (683972 [10]) \\ K(31, 2) &\geq 4464944 && (4464613 [14]) \\ K(32, 4) &\geq 110467 && (110215 [14]). \end{aligned}$$

Theorem 57. If $2 \leq R < n$, $R + 1$ divides n and

$$M = \frac{\binom{R}{2} + R - 1}{V(n, 2) - 1 - (R - 2) \left(n - \frac{R-3}{2}\right)} \leq \frac{1}{n}$$

holds, then

$$K(n, R) \geq \frac{(n - \tau)2^n}{(n - R - \tau)V(n, R) + RV(n, R - 1)}$$

with

$$\tau = (n - R) \frac{M}{M + 1}.$$

Corollary.

$$\begin{aligned} K(21, 2) &\geq 9900 && (9893 [29]) \\ K(24, 2) &\geq 60381 && (60350 [29]) \\ K(33, 2) &\geq 16210701 && (16207424 [29]) \\ K(20, 3) &\geq 892 && (889 [29]) \\ K(32, 3) &\geq 855613 && (854890 [29]) \\ K(30, 4) &\geq 38067 && (37973 [29]). \end{aligned}$$

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